



**Ministry of Health & Wellness
Jamaica**

**GUIDELINES FOR THE MANAGEMENT
OF
TUBERCULOSIS IN THE JAMAICAN
POPULATION**

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**Ministry of Health & Wellness
Jamaica**

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**GUIDELINES FOR THE MANAGEMENT OF
TUBERCULOSIS IN THE JAMAICAN POPULATION**

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Jamaica**

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GLOSSARY OF TERMS

ABC – Abacavir

ACSM - Advocacy, Communication and Social Mobilization

AFB – Acid Fast Bacilli

ATT – Anti-Tuberculosis

AZT - Zidovudine

BCG - Bacillus Calmette-Guérin

BHC – Bustamante Hospital for Children

BSL 2+ - Biosafety Level 2

CARPHA – Caribbean Public Health Agency

CBC – Complete Blood Count

CRH – Cornwall Regional Hospital

CSF – Cerebrospinal Fluid

CXR – Chest X-ray

CNS – Central Nervous System

CT- Computerised Tomography

DOPT - Directly Observed Preventative Therapy

DOT - Directly Observed Therapy

DOTS - Directly observed treatment, short course

DR-TB – Drug Resistant Tuberculosis

DST – Drug Susceptibility Testing

E - Ethambutol

EFV - Efavirenz

EPTB - Extra-Pulmonary Tuberculosis

EQAS - External Quality Assessment Schemes

FIFO - First in First Out

HAS – Hospital Active Surveillance

HRE – Isoniazid, Rifampin, Ethambutol

HSTU – HIV/STI/TB Unit

HIV/AIDS – Human Immunodeficiency Syndrome/ Acquired Immunodeficiency Syndrome

I – Isoniazid

IGRA – Interferon Gamma Release Assay

IQC – Internal Quality Control

IR – Isoniazid and Rifampin

IRP – Isoniazid, Rifampin and Pyrazinamide

IRIS - Immune Reconstitution Inflammatory Syndrome

IUATLD - International Union Against Tuberculosis and Lung Disease

LPV/r – Lopinavir/Ritonavir

LTBI – Latent Tuberculosis Investigation

MDR-TB – Multi-drug Resistant Tuberculosis

MOHW – Ministry of Health and Wellness

MOs(H) – Medical Officer(s) of Health

MOTT – Mycobacterium Other Than Tuberculosis

MRI - Magnetic Resonance Imaging

MTB – Mycobacterium Tuberculosis

MTBC – Mycobacterium Tuberculosis Complex

NAAT - Nucleic Acid Amplification Test

NCH – National Chest Hospital

NHF – National Health Fund

NNRTI - Non-Nucleoside Reverse Transcriptase Inhibitors

NPHL – National Public Health Laboratory

NPO – Nil Per Oral

NRTI - Nucleoside Reverse Transcriptase Inhibitors

NTB – National Tuberculosis

NTP - National Tuberculosis Programme

NTPCP - National Tuberculosis Prevention and Control Programme

NVP – Nevirapine

P – Pyrazinamide

PAHO – Pan American Health Organization

PHC – Primary Health Care

PI – Protease Inhibitor

PPD - Purified Protein Derivative

PPE – Personal Protective Equipment

PTB – Pulmonary Tuberculosis

R – Rifampin

RBT – Rifabutin

RHAs – Regional Health Authorities

RIPE - Rifampin, Isoniazid, Pyrazinamide, Ethambutol

RR – Rifampin Resistant

RTD – Regional Technical Director

RTV – Ritonavir

SDG – Sustainable Development Goals

SERHA – South East Regional Health Authority

SM – Streptomycin

SMO – Senior Medical Officer

SRIE – Streptomycin, Rifampin, Isoniazid, and Ethambutol

SIE - Streptomycin, Isoniazid and Ethambutol

STI – Sexually Transmitted Infection

TB – Tuberculosis

TCS – Treatment Care & Support

TST – Tuberculin Skin Test

TU - Tuberculin Units

UWI – University of the West Indies

WHO – World Health Organization

WRHA – Western Regional Health Authority

XDR-TB – Extensively Drug Resistant Tuberculosis

ZN – Ziehl Neelson

3TC – Lamivudine

FOREWARD

Tuberculosis (TB) is endemic to the Jamaican population. National statistics over the past 15 years have not indicated a resurgence in the disease occurrence. However, the concern exists that the HIV/AIDS epidemic will increase the risk factors of infection, and the number of co-infected cases will invariably increase the prevalence of the disease.

The main objective of this manual is to provide and promote standardized procedures for the prevention and control of TB in Jamaica. Successful implementation of the policy package is dependent largely on the commitment of health professionals at all levels in the health sector.

It will serve as a management tool, equipping health professionals with the necessary technical resource to effectively manage any eventualities they may encounter in case management. Establishment of the new manual will also satisfy the requisition by the World Health Organization (WHO) global TB strategy, for member countries to provide a procedure manual that will serve as a management tool and as a training resource manual. The manual will emphasize the modern strategy for treatment as recommended by the WHO and will incorporate the short course chemotherapeutic regime as applied under the directly observed treatment, short course (DOTS) management strategy. This is in keeping with the key strategy of any National Tuberculosis Programme (NTP); which is the early detection and cure of all infectious cases.

The preparation of the manual included resource material from:

- Caribbean Guidelines for the Prevention, Treatment, Care and Control of Tuberculosis and TB/HIV (Third Edition, January 2010)
- Guidelines on the management of latent tuberculosis infection (World Health Organization, 2015)
- Latent tuberculosis infection: Updated and consolidated guidelines for programmatic management (World Health Organization, 2018)
- Definitions and reporting framework for Tuberculosis - 2013 revision (updated December 2014. World Health Organization; 2013)
- Guidance for national tuberculosis programmes on the management of tuberculosis in children 2nd Edition (World Health Organization, 2014)
- Expertise of local health professionals

I. POLICIES, STRUCTURES, GOALS AND OBJECTIVES

I. POLICIES, STRUCTURES, GOALS AND OBJECTIVES

1.1 Policies

1.2 Structures

1.3 Epidemiology

1.4 Targets

1.5 Objectives

1.6 Key Strategies

1.1 Policies

Jamaica is a party to several TB related resolutions endorsed at the different sittings of the PAHO Directing Council and the World Health Assembly. The management of the National Tuberculosis Programme lies within the HIV/STI/TB Unit of the Ministry of Health and Wellness, and receives oversight from the Health Promotion and Protection Branch.

The TB policy and procedure manual is developed to provide a standardized approach to the management of TB in Jamaica. It is in line with the National Strategic Plan for Tuberculosis 2020 – 2024 goal, targets and objectives; and the guiding principles of the World Health Organization END TB Strategy, adapted for the Jamaican context.

TB is classified as a Class 1 Notifiable Disease, governed under the Public Health (Class 1 Notifiable Diseases) Order, 2003. Accordingly, the case must be reported within twenty-four hours of being discovered to the parish Public Health Department or the National Surveillance Unit of the Ministry of Health and Wellness.

The National Chest Hospital (NCH), the Bustamante Hospital for Children (BHC) and the Cornwall Regional Hospital (CRH) serve as the primary diagnostic and management centers for TB cases in Jamaica.

1.2 Structures

The National Tuberculosis Prevention and Control Programme (NTPCP) is managed at three levels:

1. Central level – policies, guidelines, regulations, monitoring and evaluation
2. Regional level – supervision of operations
3. Parish level – implementation of policies and guidelines

Central Level

The management of the National Tuberculosis Programme at the central level involves the National Public Health Laboratory, the HIV/STI/TB and National Surveillance Units of the Ministry of Health and Wellness; and the National Health Fund.

The central level develops, implements, monitors, and evaluates all aspects of the TB programme including development of guidelines, monitoring and evaluation and the procurement of drugs and laboratory supplies. Drugs are procured and distributed by the National Health Fund (NHF) and diagnostic services provided by the National Public Health Laboratory.

Regional Level

The Regional Health Authorities (RHAs) are charged with the supervision of the Parish operations within the RHAs and serve as the connection between the National and Parish levels.

Parish Level

The Parish level is responsible for the implementation of the policies and guidelines and ensuring the activities of the National Tuberculosis Programme are delivered at the end user level. This involves case detection, referral for and completion of treatment and investigation of contacts.

1.3 Epidemiology

Jamaica is currently classified by the World Health Organization (WHO) as a low-burden (less than 10 cases per 100,000 population per year) TB country with an estimated incidence of 2.9 per 100,000 population (n=80) in 2018 (Source: NSU). During the five-year period 2014 -2018, there were 544 cases of Tuberculosis diagnosed (confirmed) and treated in Jamaica, an average of 109 cases per annum. Of the confirmed cases of Tuberculosis for the period, most (58%) resided in the South East Regional Health Authority (SERHA) and 28.6% resided in the area administered by the Western Regional Health Authority (WRHA). Together, the SERHA and WRHA consistently contributed over 80% of the confirmed tuberculosis cases identified in the 5-year period.

In 2018, there were 80 cases of confirmed TB in Jamaica, of these 59 were male and 21 female (2.8:1), consistent with the global TB picture of a greater incidence in males than females (2:1). The largest number of cases was detected in adults between 25 and 34 years old (n=16; 20%), and this was consistent across the 5-year span between 2014 and 2018.

Of the cases confirmed in 2018, 3.8% (n=3) were hypertensive, 2.5% (n=2) diabetic and 1.25% (n=1) had sickle cell disease and asthma respectively.

The WHO defines a TB death as a patient who died on treatment for any reason before or during the course of treatment. Adequate disease control measures are expected to result in a case fatality rate of 5% or less. In 2018, 4 (5.0%) persons diagnosed with TB died while on treatment.

1.4 Goal

The Goal of the Tuberculosis Programme is to reduce the incidence and mortality from Tuberculosis, thereby accelerating the elimination of Tuberculosis in Jamaica; no family should face catastrophic costs due to Tuberculosis.

The achievement of this goal is through the adaptation of the action framework and guiding principles from the World Health Organization (WHO) END TB Strategy for low incidence countries. The priority action areas include:

- 1) Ensure political commitment, funding and stewardship for planning and essential services of high quality
- 2) Address the most vulnerable and hard to reach groups
- 3) Address special needs of migrants and cross-border issues
- 4) Undertake screening for active TB and Latent Tuberculosis Infection (LTBI) in TB contacts and selected high-risk groups and provide appropriate treatment
- 5) Optimize the prevention and care of drug-resistant TB
- 6) Ensure continued surveillance, programme monitoring and evaluation and case-based data management
- 7) Invest in research and new tools
- 8) Support global TB prevention, care and control

1.5 Targets of the National Tuberculosis Programme include:

Targets to be achieved by 2025:

1. 75% reduction in Tuberculosis deaths (compared with 2015)
2. 50% reduction in Tuberculosis incidence rate (compared to 2015)
3. No affected families facing catastrophic costs due to Tuberculosis
4. To maintain case detection rates of ≥ 85 percent of expected cases
5. To achieve a cure rate of ≥ 90 percent of confirmed cases

Targets to be achieved by 2035:

1. 95% reduction in TB deaths (compared to 2015)
2. 90% reduction in TB incidence rate (compared to 2015)
3. No affected families facing catastrophic costs due to TB.

1.6 Key Strategies

The key strategies of the Tuberculosis Programme are aimed to:

1. Reduce morbidity and mortality associated with the disease by decreasing or interrupting tuberculosis transmission in Jamaica.
2. Cure the disease in affected individuals, to preserve or quickly restore their work capacity allowing them to remain with their family and community and in this way, maintain their socio-economic position.
3. Decrease the risk of tuberculosis infection in the community and by this means, improve the control situation of tuberculosis, and thus the health, social and economic condition of the community.

To achieve this there are three Strategic Interventions with specific strategic actions as outlined below.

1. Integrated Patient-Centered Care and Prevention of TB

Strategic Actions:

- 1.1 Revamp the NTP and strengthen its capacity for management and technical programming.
- 1.2 Rapidly increase early and appropriate diagnosis of TB.
- 1.3 Implement vigorous advocacy, communication and social mobilization (ACSM).
- 1.4 Implement active case finding activities in vulnerable populations.
- 1.5 Ensure timely and appropriate management of drug-susceptible and drug-resistant TB.
- 1.6 Strengthen the integration of TB and HIV care to optimize the management of co-infected patients.
- 1.7 Promote joint management of other co-morbidities associated with TB.
- 1.8 Review and strengthen the management of latent TB infection.

2. Promote Universal Access to Care for TB

Strategic Actions:

- 2.1 Increase allocation of human and financial resources to the TB control programme.
- 2.2 Promote collaboration with civil society for TB prevention and control.
- 2.3 Optimize TB data management including surveillance and vital registration.

2.4 Ensure steady supply of anti-TB drugs and laboratory commodities.

2.5 Inclusion of TB in social protection systems, poverty reduction programmes, and action on social determinants of TB.

3. Promote Operational Research and Implementation of Innovative Initiatives and Tools for TB Control

Strategic Actions:

3.1 Review new diagnostic tools and medicines.

3.2 Strengthen monitoring and evaluation as a platform for evidence-based programming.

3.3 Implement operational research to generate evidence on the TB situation and for TB control programme.

Guideline: Policies, Structures, Goals and Objectives		
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CHAPTER: II

TUBERCULOSIS TRANSMISSION AND PATHOGENESIS

CHAPTER: II

TUBERCULOSIS TRANSMISSION AND PATHOGENESIS

2.1 Aetiology of Tuberculosis

2.2 Transmission of Tuberculosis

2.3 Pathogenesis of Tuberculosis

2.4 Clinical Features

2.5 Diagnosis

2.6 Classification of Persons Exposed to or Infected
with *Mycobacterium Tuberculosis*

2.1 Aetiology of Tuberculosis

TB is caused by the *Mycobacterium tuberculosis* complex. These appear as acid-fast bacilli identified using Ziehl Neelson staining. The complex includes *Mycobacterium tuberculosis* and *M. africanum* which have human reservoirs, and *M. bovis* from cattle and infected, unpasteurised milk. Other mycobacteria like *M. avium complex* and *M. kansasii* can produce disease clinically indistinguishable from tuberculosis. These mycobacteria other than TB (MOTT), can be identified only by culture of the organisms.

2.2 Transmission of Tuberculosis

TB in the lungs or throat can be infectious, that is, bacteria can be spread from person to person. TB in other parts of the body such as the kidney or spine is usually not infectious.

TB is spread from person to person through the air by droplet nuclei of 1 -5 microns in diameter produced when persons with pulmonary or laryngeal tuberculosis disease cough, sneeze, speak or sing. These small particles contain the *M. tuberculosis complex* and depending on the environment may remain suspended in the air for several hours and can be inhaled by the susceptible exposed individual. The smaller particles reach the alveoli within the lung where the organisms replicate. Larger particles do not remain airborne for long and if inhaled do not reach the alveoli but are trapped in mucus and expectorated or swallowed. Tubercle bacilli deposited on intact mucosa or skin does not invade the exposed individual.

The likelihood of transmission of *M. tuberculosis complex* is determined by:

1. The number of organisms being expelled into the air
2. The volume of space and the ventilation
3. The length of time an exposed individual breathes the contaminated air
4. The immune status of the exposed individual

2.3 Pathogenesis of Tuberculosis

After inhalation, the droplet nucleus implants in the alveolus. It multiplies within the alveolar macrophage, growing slowly and producing no immediate host response. The organisms grow for 2 to 12 weeks until they reach numbers which will elicit a cellular immune response that can be detected by a reaction to the tuberculin skin test (TST). Multiplication of the tubercle bacillus takes place in the alveoli, giving rise to an infection known as the Ghon focus, which is a primary lesion caused by tubercle bacilli in the lungs. It is a small area of granulomatous inflammation, only detectable by chest X-ray (CXR) if it calcifies or grows substantially; typically, these will heal.

Before the development of cellular immunity tubercle bacilli spread via the lymphatics to the ipsilateral lymph nodes and then through the bloodstream to more distant sites. The Ghon focus, with associated hilar adenopathy is known as the Ghon complex. In persons with intact cellular immunity, collections of activated T cells and macrophages form granulomas that limit multiplication and spread of the organism. However, the bacilli may remain dormant for years. At this stage the majority of pulmonary tuberculosis infections are clinically and radiographically undetectable. A positive tuberculin skin test being the only indication that infection with *M. tuberculosis* has taken place. Individuals with latent tuberculosis infection but no active disease are not infectious and cannot transmit the organism. Ten percent (10%) of immunocompetent hosts with tuberculosis infection will progress to disease. Fifty percent (50%) of these cases occur 1-2 years after infection.

Risk factors for progression from infection to TB disease

There are certain conditions that reduce the ability of the individual to respond to the pathogen. These include host factors:

- Prior history of active TB, particularly in cases of inadequate treatment
- Children under 5 years of age and the elderly
- Substance abuse (illicit drugs and alcohol)
- Tobacco smoking

Medical conditions which increase risk include:

- HIV co-infection
- Silicosis
- Diabetes mellitus
- Severe liver disease
- Chronic renal failure
- Solid organ transplantation
- Gastrectomy or jejunioileal bypass

- Certain types of cancer such as leukemia and Hodgkin's disease
- Malnutrition or low body weight (<10% below normal)

The use of immunosuppressive drugs is also a risk factor. These include:

- Corticosteroids for more than 4 weeks in doses >15mg
- Cancer chemotherapy drugs

HIV infected persons especially those with low CD4 cell counts develop tuberculosis disease rapidly after becoming infected with *M. tuberculosis*; up to 50% of such persons do so within 2 years of infection. Those individuals with latent tuberculosis who acquire HIV infection will develop tuberculosis at a rate of 5-10% per year. There is also a higher likelihood of haematogenous spread and disseminated disease.

2.4 Clinical Features

The clinical presentation of tuberculosis is quite variable. There are two major types of tuberculosis: pulmonary (85%) and extra-pulmonary (15%).

Pulmonary Tuberculosis

Pulmonary tuberculosis (PTB) refers to any bacteriologically confirmed or clinically diagnosed case of TB involving the lung parenchyma or the tracheobronchial tree. Miliary TB is classified as PTB because there are lesions in the lungs. Tuberculous intra-thoracic lymphadenopathy (mediastinal and/or hilar) or tuberculous pleural effusion, without radiographic abnormalities in the lungs, constitutes a case of extra-pulmonary TB. A patient with both pulmonary and extra-pulmonary TB should be classified as a case of PTB (World Health Organization [WHO], 2014).

Pulmonary infection is most often manifested by:

- Chronic cough (> 2 weeks): non-productive in early disease, but producing sputum as inflammation and tissue necrosis occur
- Haemoptysis is usually the result of previous disease
- Chest pain may result from pleuritic pain
- Dyspnea is usually only found in extensive disease

Physical findings are non-specific, with occasional crackles in the area of involvement and bronchial breath sounds if there is lung consolidation.

Extra-Pulmonary Tuberculosis

Extra-pulmonary tuberculosis (EPTB) refers to any bacteriologically confirmed or clinically diagnosed case of TB involving organs (World Health Organization, 2014) other than the lungs, e.g. lymph nodes, pleura, genitourinary tract, skeletal, central nervous system, abdomen, pericardium and skin (WHO, 2014). Extra-pulmonary tuberculosis is more difficult to diagnose than pulmonary tuberculosis, because it is less common and involves relatively inaccessible sites, where fewer bacilli can cause much greater damage.

Tuberculosis involving any site may produce symptoms and signs that are not specifically related to the organ or tissue involved, but rather which are systemic in nature. These symptoms and signs include:

- Fever
- Anorexia
- Weight loss
- Malaise

- Night sweats
- **Lymph Node Tuberculosis**

Tuberculous lymphadenitis usually presents as painless swelling of one or more lymph nodes. The nodes most commonly involved are those of the posterior or anterior cervical chain or supraclavicular fossa. The nodes are initially discrete and overlying skin is normal. With continuing disease, the nodes may become matted and overlying skin inflamed. Intrathoracic nodal involvement may compress bronchi causing atelectasis and lung infection with bronchiectasis. This manifestation is common in children.
- **Pleural Tuberculosis**

This commonly goes unnoticed and resolves spontaneously. In some patients, pleural tuberculosis is manifested as an acute illness with fever and pleuritic pain. If there is a large effusion, dyspnea may occur. Pulmonary disease is nearly always present. Involvement of the pleura may also result from rupture of an adjacent cavity, and may lead to emphysema.
- **Genitourinary Tract Tuberculosis**

Local symptoms predominate. These include dysuria, haematuria, frequency of urination, and flank pain. However, symptoms may be subtle and there may be advanced destruction of kidneys by the time a diagnosis is made. In women, pelvic pain, menstrual irregularities and infertility may be the presenting complaints of genital involvement. In men a slightly painful scrotal mass is the most common presenting symptom of genital involvement, but symptoms of prostatitis, orchitis and epididymitis may occur. Many patients are asymptomatic and detected through an abnormal urinalysis, which may show pyuria or haematuria.
- **Skeletal Tuberculosis**

This usually presents with pain. Swelling of the joints may be noted, associated with limitation of movement and sinus tract formation. Skeletal tuberculosis is more common in children than in adults. Computerised Tomography (CT) scan and Magnetic Resonance Imaging (MRI) are more sensitive than routine films. A bone biopsy may be needed to obtain diagnostic material.
- **Central Nervous System (CNS) Tuberculosis**

Tuberculous meningitis can result from direct seeding or an old para-meningeal focus with rupture into the subarachnoid space. The process is located primarily at the base of the brain leading to cranial nerve involvement, headache, decreased

level of consciousness and neck stiffness. This is usually investigated by lumbar puncture, CT scan or MRI.

- **Abdominal Tuberculosis**

This may occur in any abdominal organ as well as the peritoneum; clinical manifestations depend on the areas of involvement. The most common sites are the terminal ileum and caecum where it may cause pain and intestinal obstruction. A palpable mass and abnormal barium enema may be mistaken for carcinoma. Rectal lesions may present as fissures, fistulae or abscesses. Tuberculous peritonitis frequently causes pain as its presenting manifestation, accompanied by abdominal swelling, ascites and fever.

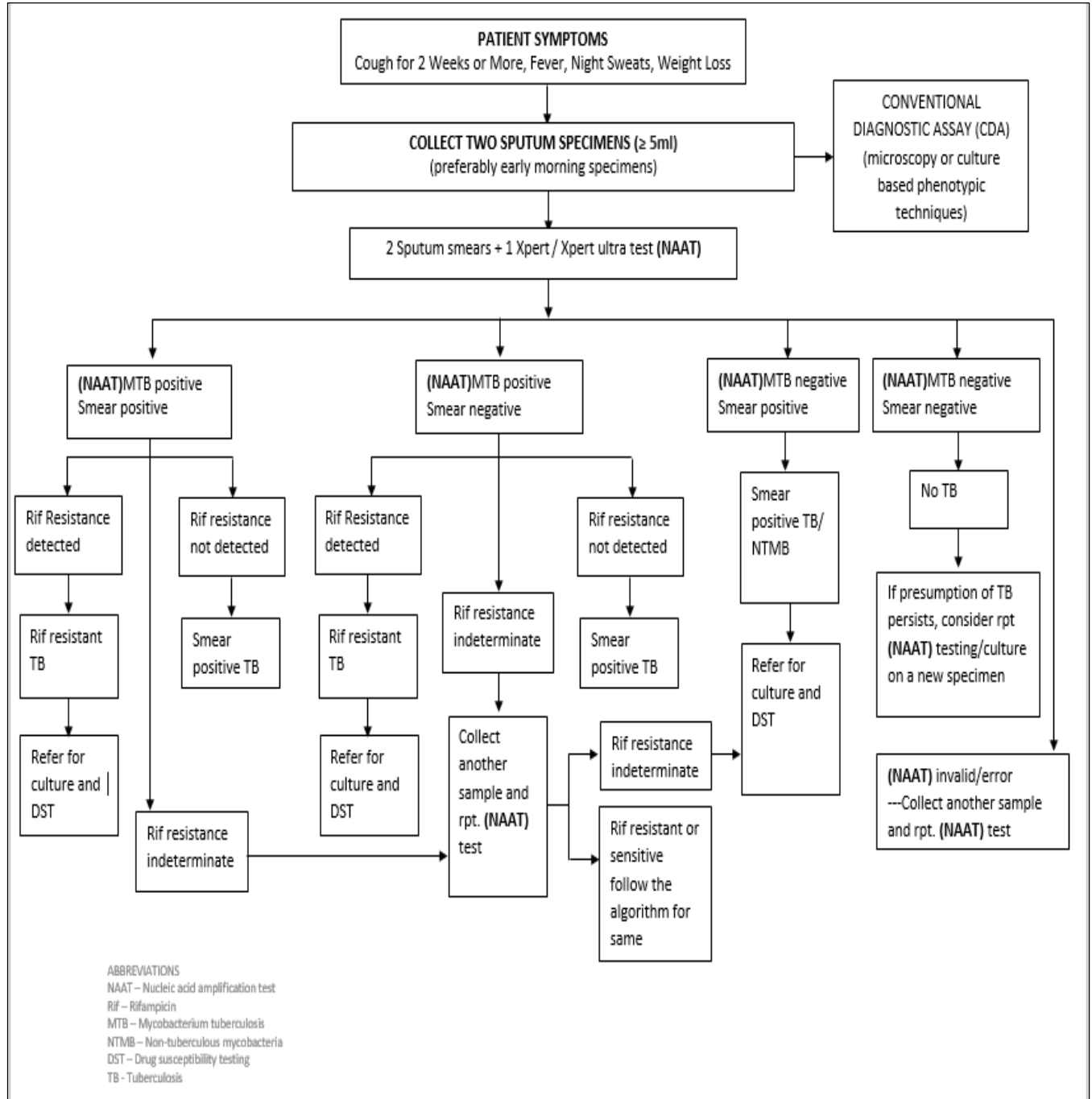
- **Pericardial Tuberculosis**

This may cause pain, effusion, and haemodynamic effects. The systemic symptoms are non-specific. Cough, dyspnea, orthopnea, ankle swelling and chest pain tend to occur later.

2.5 Diagnosis

Figure 2.5.1 shows the clinical algorithm for the investigation and diagnosis of tuberculosis. Any person presenting with a cough of more than 2 weeks duration should be investigated for tuberculosis.

Figure 2.5.1: Clinical Algorithm for the Investigation and Diagnosis of Tuberculosis



This investigation consists of the following:

1. A history and physical examination with a focus on:

- Exposure to a person who has infectious TB
- Symptoms of TB
- Past TB infection or disease
- Risk factors/Co-morbid conditions

2. The tuberculin skin test (Mantoux) is usually positive

This is performed by injecting 0.1 ml of 5 tuberculin units (TU) of purified protein derivative (PPD) intra-dermally, usually on the forearm. The skin is stretched with the thumb, needle inserted with line and bevel outwards, keeping the needle almost parallel to the skin, the needle is pushed 2-3mm and PPD injected. A blanched wheal should appear in which the skin follicles are readily discernible.

The patient's arm is examined 48 to 72 hours later for induration. The diameter of the indurated area across the forearm is measured. The area of erythema around the induration does not indicate infection and should not be measured. The TST may be interpreted as follows:

- An induration ≥ 5 mm is classified as positive in the following groups:
 - Immunosuppressive conditions, HIV, drugs (e.g. chemotherapeutic agents, etc.
 - Radiographic evidence of active or previously active TB
- A tuberculin reaction of ≥ 10 mm of induration is classified as positive in persons who have other risk factors for TB including:
 - Recent arrivals from high prevalence countries
 - Frequent exposure to high risk individuals
 - Residents and employees of high risk congregate settings: prisons, nursing homes etc.
 - Children under 4 years of age
 - Medical conditions such as lymphoma, diabetes mellitus, chronic renal failure and malnutrition
 - Recent contacts of persons diagnosed with TB
- A tuberculin reaction of ≥ 15 mm of induration is classified as positive in persons:
 - With no known risk of TB
 - Age 4 years and older.

Persons who have received BCG vaccine are likely to have an induration ≥ 10 mm if tested less than 15 years after vaccination. If BCG vaccination was given more than 15 years previously, it should be ignored as a cause of a current positive skin

test result especially if the induration is greater than 15mm. Large or strongly positive skin tests are most probably due to TB infection rather than BCG.

3. Bacteriological Investigation

Sputum or specimen appropriate to the affected site should be collected and transferred to the TB Laboratory at the National Public Health Laboratory (NPHL).

4. Radiological Investigation

Chest X-ray should be done on all patients. There are no chest x-ray findings that are pathognomonic for tuberculosis. There are varying findings such as: cavitory disease, consolidation, pleural effusions, and fibrotic changes. A high index of suspicion must be maintained.

5. HIV Testing

An HIV test must be done on each patient at the time of TB diagnosis. Patients with documentary evidence of confirmed HIV infection are exempted, but should be counted among TB patients tested for HIV.

2.6 Classification of Persons Exposed to or Infected with *Mycobacterium Tuberculosis*

1. No tuberculosis exposure: not infected. There is no history of exposure and a negative tuberculin skin test.

2. Tuberculosis exposure: no evidence of infection. There is a history of exposure, but a negative reaction to the skin test. A follow up test should be done 6 to 8 weeks after the last exposure.

3. Latent tuberculosis: no active disease. There is a positive skin test; negative radiographic, bacteriological and clinical findings. Treatment may be indicated for some persons in this group. The person may have had a previous episode of tuberculosis, or an abnormal radiographic finding and now has a positive skin test, but no other evidence of disease. These persons may never have received chemotherapy, or may be receiving treatment, or may have completed a course of chemotherapy (See Chapter 6: Latent Tuberculosis Infection).

4. Active tuberculosis: All patients with clinically active tuberculosis whose diagnostic procedures are complete. A person must have clinical, bacteriological and/or radiographic evidence of current tuberculosis. A person remains in this class until the treatment is completed.

5. Presumptive tuberculosis case: A patient who has strong clinical or radiological findings in keeping with TB and are pending laboratory diagnosis. Persons should not remain in this class for more than 3 months.

Treatment

Standardized treatment regimens exist in keeping with the type and classification of TB being treated.

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CHAPTER: III

MANAGEMENT OF TUBERCULOSIS IN ADULTS

CHAPTER: III

MANAGEMENT OF TUBERCULOSIS IN ADULTS

3.1 Aims of Treatment

3.2 Standardized Treatment Regimen

3.3 Recommended Treatment Regimens for Different
Treatment Categories

3.4 Management of Chronic and Multi-Drug-Resistant
(MDR-TB) Cases

3.1 Aims of Treatment

The aims of treatment of tuberculosis are:

- To prevent death from active TB or its late effects
- To cure the patient of TB
- To prevent relapse of TB
- To decrease TB transmission to others

Anti-TB treatment will achieve these aims and prevent the emergence of drug resistant *M. Tuberculosis*.

There are three main properties of anti-tuberculosis drugs:

1. Bactericidal activity
2. Sterilizing activity
3. Ability to prevent resistance

The classes of anti-TB drugs are divided into first and second line anti-TB drugs. First line anti-TB drugs are:

1. **Isoniazid** is bactericidal, killing 90% of the total population of bacilli during the first few days of treatment.
2. **Rifampin** is bactericidal.
3. **Pyrazinamide** is bactericidal.
4. **Ethambutol** is bacteriostatic and used in association with the bactericidal drugs to prevent emergence of resistant bacilli.

3.2 Standardized Treatment Regimen

The standard treatment regimen involves the use of four antibiotics Rifampin, Isoniazid, Pyrazinamide and Ethambutol (RIPE). The combination therapy has proven to be effective against drug sensitive Mycobacterium Tuberculosis Complex (MTBC). Monotherapy is contraindicated.

Table 3.2.1 shows the daily dose and dosage range for anti-TB first line drugs used locally.

Table 3.2.1: Dosage of Anti-TB First Line Drugs

		Daily dose and range (mg/kg body weight)	Maximum daily dose (mg)
Rifampin (R)	Bactericidal	10 (8-12)	600
Isoniazid (I)	Bactericidal	5 (4-6)	300
Pyrazinamide (P)	Bactericidal	25 (20-30)	2000
Ethambutol (E)	Bacteriostatic	15 (15-20)	—

N.B Second line anti-Tb drugs are used to treat drug resistant TB.

Tuberculosis Treatment Regimens

The treatment of Tuberculosis has an intensive phase lasting a minimum of 2 months and a continuation phase lasting 4-6 months.

Intensive or Initial Stage

During the intensive phase, treatment consists of the four drugs RIPE. Many patients will become non-infectious within two weeks and sputum smear positive TB become smear negative within 6-8 weeks.

Directly observed therapy (DOT) is essential during treatment. This improves compliance and helps to prevent drug resistance.

Continuation or Consolidation Stage

During the continuation phase, usually only rifampin and isoniazid are required. Directly observed therapy should be continued during the consolidation phase.

The recommendation is for daily directly observed therapy (DOT).

3.3 Recommended Treatment Regimens for Different Treatment Categories

Table 3.3.1 shows the treatment regimens by TB treatment category.

Table 3.3.1: Treatment Regimens by TB Treatment Category

TB Treatment Category	Case Clinical Presentation	Treatment Regime	
		Initial Phase (Daily)	Continuation Phase (Daily)
New and other patients with sensitive TB	- Smear +/- PTB - All forms of EPTB except TB meningitis and Osteoarticular TB	2 Months RIPE	4 months IR
	TB Meningitis and Osteoarticular TB	2 Months RIPE	10 months IR
Drug resistant TB (DR-TB) patients other than RR/MDR-TB		Regimen according to the drug sensitivity testing (DST)	
Presumptive RR/MDR TB Case	Contacts of RR/MDR-TB patients. RR/MDR	Same Regimen as index case	
Confirmed RR case by any DST method		MDR-TB regimen plus Isoniazid	
Confirmed MDR-TB case		MDR-TB standardized regimen	

RR- Rifampin Resistant; DST – Drug Sensitivity Testing

Special Considerations

Pregnant women

The first line anti-TB drugs are safe for use in pregnant women; except for streptomycin, which is ototoxic to the foetus and should not be used in pregnancy.

Women on oral contraceptives

Rifampin interacts with the oral contraceptive pill and can decrease efficacy against pregnancy. The contraceptive method should be reviewed by the physician.

HIV infection

All suspected or confirmed TB patients should be tested for HIV infection after receiving appropriate counseling. The standard anti-TB regimen will have to be modified depending on the ARV therapy and the severity of the disease in patients with HIV. **Specialist referral will be needed.**

Patients with liver disorders

Patients with hepatitis virus carriage, a past history of acute hepatitis, or excessive alcohol consumption can receive the standard anti-TB regimens provided that there is no clinical evidence of chronic liver disease. These patients need to be monitored closely. Clinical monitoring and liver function tests should be performed during treatment in all patients with pre-existing liver disease. Expert consultation should be sought when treating TB patients with advanced or unstable liver disease.

Established chronic liver disease

Patients with liver disease-should not receive pyrazinamide. The recommended regimens are the following:

- 1) Initial phase of 2-months daily SRIE followed by 6 months daily IR; or
- 2) 2-months daily SIE followed by 10 months daily IE.

Specialist referral will be needed.

Acute hepatitis (e.g. acute viral hepatitis)

In cases where it is necessary to treat TB during acute hepatitis that is unrelated to TB or anti TB treatment, the combination of streptomycin and ethambutol up to a maximum duration of 3 months is the safest option until the hepatitis has resolved. The patient can then receive a continuation phase of 6 months isoniazid and rifampin. Clinical judgment is necessary to determine if TB treatment should be given or deferred until hepatitis has resolved.

Patients with renal failure

Isoniazid, rifampin and pyrazinamide can be given in the normal dosage to patients with renal failure; these drugs are either eliminated almost entirely by biliary excretion or metabolized into non-toxic compounds. Ethambutol is excreted by the kidney and should be avoided. The recommended regimen to be administered to patients with renal failure

is 2 months daily IRP followed by 6 months daily IR. In severe renal failure, patients should receive pyridoxine with isoniazid in order to prevent peripheral neuropathy.

Six-month treatment regimen

Intensive phase - 2 months - Rifampin

Isoniazid

Pyrazinamide

Ethambutol

Continuation phase- confirm sputum negativity - Rifampin

Isoniazid

Interim reports as detailed in the Case Investigation Form (Appendix I) must be sent to the relevant public health department (Medical Officer of Health) at the end of the intensive and continuation phases.

3.4 Management of Chronic and Drug-Resistant Tuberculosis

Cases are classified as detailed below based on drug susceptibility testing (DST) of clinical isolates confirmed to be *M. Tuberculosis*. These categories are not mutually exclusive.

- **Mono-resistance:** *M. tuberculosis* found to be resistant to one of the first line anti-tuberculosis (ATT) drugs only.
- **Poly-resistance:** Bacteria are resistant to more than one of the ATT drugs (other than both isoniazid and rifampin).
- **Multidrug resistant TB (MDR-TB):** Bacteria resistant to at least both isoniazid and Rifampin.
- **Extensive drug resistant TB (XDR-TB):** Resistance to any fluoroquinolone and to at least one of three second line injectable drugs (capreomycin, kanamycin and amikacin) in addition to multi-drug resistance.
- **Rifampin resistance:** Resistance to rifampin detected using phenotypic or genotypic methods, with or without resistance to other anti-TB drugs. It includes any resistance to rifampin, whether mono-resistance, multidrug resistance, polydrug resistance or extensive drug resistance.

Background

MDR-TB is rarely observed in new cases. The risk is increased and is more frequent in re-treatment cases, especially in failure cases.

The aims of treatment in MDR-TB cases are similar to those of all cases with TB. However, MDR-TB patients respond poorly to short-course chemotherapy and need to be treated intensely and for up to 24 months with a regimen based on second line anti-tuberculosis drugs.

Full implementation of directly observed therapy DOT (Chapter 9) is the best prevention against MDR-TB.

Managerial Principles

Management of MDR- TB cases should be carried out by the specialist institution (NCH) and cases must be isolated.

The main priority for MDR-TB control is the identification and cure of sputum smear-positive cases. The decision to use second line anti-tuberculosis drugs should be based on:

- The availability of financial resources
- Laboratories that can perform high-quality drug susceptibility testing
- Prevention of uncontrolled use of second line drugs

- Special notification and registration of chronic and MDR-TB cases
- Convening of expert committee(s) for decisions on treatment and monitoring. (Proposed membership of this committee to include National Chest Hospital (NCH) Consultant/SMO, NTP Manager, Pulmonologist, Paediatrician, Infectious Disease Consultant, MOHW Epidemiologist, Pharmacist and PAHO Technical Advisor)

Treatment of MDR-TB cases with second line drugs is more expensive and more toxic than treatment with essential drugs. Hospitalization entails increased risk of nosocomial transmission of MDR-TB to both staff and patients, especially those infected with HIV.

After tolerance of the drug regimen has been ascertained and the patient's cooperation has been secured, the patient can be started on ambulatory treatment. Ambulatory treatment reduces the risk of MDR-TB transmission in hospitals.

Management of MDR-TB cases with second line drugs can be done using standardized or individualized regimens. These will be determined based on the recommendations of the expert committee.

Special efforts are needed to persuade patients to complete the long treatment regimens required. The advantages of individualized regimens include treatment according to the susceptibility pattern and, likely, higher cure rates.

This approach may, however, be costlier than standardized regimens in terms of the drugs involved, in laboratory capability and in the training required to administer a variety of treatment regimens. Standardized treatment regimens have been proven to be successful and are useful in settings where second line drugs have not been used extensively and hence resistances to these drugs are low.

Principles of Treatment

The treatment regimen should include at least 4 drugs, including an injectable agent and a fluoroquinolone in the initial phase, and at least 3 of the most active and best-tolerated drugs in the continuation phase. An initial phase of at least 6 months should be followed by a continuation phase of 12-18 months.

Samples that require drug susceptibility testing should be routed through the NPHL which can liaise with international laboratories.

Reserve Anti-Tuberculosis Drug

Second line drugs are most useful for the management of MDR-TB cases. Table 3.4.1 shows the second line anti-tuberculosis drugs, their mode of action and recommended dosage. Table 3.4.2 shows the suggested treatment regimens for MDR-TB.

Table 3.4.1: Reserve Anti-Tuberculosis Drugs

Reserve Abbreviation	Drug	Mode of Action	Recommended Adult Daily Dosage ¹		
			Average (MG/KG)	Minimum (MG)	Maximum (MG)
Amikacin (Am)		Bactericidal	15	750	1000
Capreomycin (Cm)		Bactericidal	15	750	1000
Ciprofloxacin (Cx)		Bactericidal	10-20	1000	1500
Ethionamide (Et)		Bactericidal	10-20	500	750
Kanamycin (Km)		Bactericidal	15	750	1000
Ofloxacin (O)		Bactericidal	7.5 – 15	600	800
P-Aminosalicylic Acid (PAS)		Bacteriostatic	150	8g	12g
Protionamide (Pt)		Bactericidal	10-20	500	750

¹Thrice weekly regimens are not recommended

Table 3.4.2: Suggested Treatment Regimens for MDR-TB

Susceptibility Testing to Essential Drugs	Initial Phase		Continuation Phase	
	Drugs	Duration	Drugs	Duration
Not available ²	Km ³ + Et + O ⁴ + P + 1-E	At least 6 months	Et + Q + P +/- E	12 – 18 months
Available: Resistance to I + R	S ⁵ + Et + Q ⁶ + P +/- E	At least 6 months	Et + Q + P +/- E	12 – 18 months
Resistance to all essential drugs	1 injectable + 1 fluoroquinolone + 2 of these oral drugs: PAS, Et, Cs	At least 6 months	The same drugs except injectable	18 months
Susceptibility testing to reserve drugs	Tailor regimen according to susceptibility pattern			

AM- Amikacin, Cm – Capreomycin, Cx- Ciprofloxacin, Et – Ethionamide, Km – Kanamycin, O –Ofloxacin, PAS – P-Aminosalicylic Acid, Pt – Protionamide

P –Pyrazinamide, E –Ethambutol, I –Isoniazid

²Use of a standardised - regimen could be feasible in resource-limited countries with a high burden of TB and a strong and efficient NTP.

³Am or Cm could also be used. However, since there is cross-resistance between Km and Am, if either drug was used previously or if resistance to them is suspected, Cm is the preferred choice.

⁴Fluoroquinolone (ciprofloxacin or ofloxacin).

⁵If resistance to S is confirmed, replace this drug with Km, Am or Cm.

⁶Individualised regimen is probably more feasible in designated centres of excellence.

In December 2019, the World Health Organization disseminated the document Rapid Communication: Key changes to the treatment of drug-resistant tuberculosis (WHO, 2019). Key points from this document are:

- All patients with MDR/RR-TB, including those with additional resistance to fluoroquinolones, stand to benefit from effective all-oral treatment regimens, either shorter or longer, implemented under programmatic conditions.
- Decisions on appropriate regimens should be made according to patient preference and clinical judgement, also considering the results of susceptibility testing, patient treatment history and severity and site of the disease.
- For MDR/RR-TB patients without previous exposure to second-line treatment (including bedaquiline), without fluoroquinolone resistance and no extensive TB disease or severe extrapulmonary TB, the preferred treatment option is a shorter, all-oral bedaquiline-containing regimen. Access to rapid drug susceptibility testing, especially for ruling out fluoroquinolone resistance is required before starting the shorter, all-oral, bedaquiline-containing MDR-TB regimen.

Bedaquiline is a bactericidal drug which belongs to a new class of antibiotics (diarylquinolines). The mechanism of action is by interference with the function of an enzyme required by the tuberculosis bacterium to produce energy and to replicate. Currently bedaquiline is not used in Jamaica. The NTP will determine the transition to this treatment regimen in the local context and update the national guidelines as necessary.

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CHAPTER: IV

MANAGEMENT OF TUBERCULOSIS IN CHILDREN

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MANAGEMENT OF TUBERCULOSIS IN CHILDREN

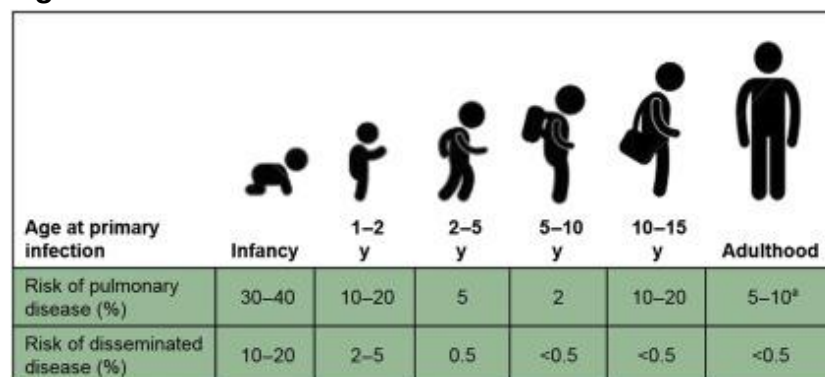
- 4.1 Pathogenesis of TB in Children
- 4.2 Clinical Features
- 4.3 WHO Case Definitions
- 4.4 Diagnosis of TB Disease in Children
- 4.5 Standardized Treatment Regimen
- 4.6 Special Populations
- 4.7 Breastfeeding Women
- 4.8 BCG Vaccine

4.1 Pathogenesis of TB in Children

Infection with *M. tuberculosis* complex usually results from inhalation into the lungs of aerosolized infected droplets produced by someone who has pulmonary or laryngeal TB. The source of infection of most children is an infectious adult in their close environment (usually the household). This exposure leads to the development of a primary parenchyma lesion (Ghon focus) in the lung with spreading to the regional lymph node(s). The immune response (delayed hypersensitivity and cellular immunity) develops about 4-12 weeks after the primary infection. In most cases, the immune response stops the multiplication of *M. tuberculosis* bacilli at this stage. However, a few dormant bacilli may persist. A positive tuberculin skin test (TST) would be the only evidence of infection.

Disease occurs when the immune response is not strong enough to contain the infection. Inhaled bacteria which are not destroyed result in a primary infection consisting of a small parenchymal focus which spreads locally to regional lymph nodes. The risk of disease progression follows a bimodal pattern as shown in Figure 4.1.1. Infants and young children under age 4 years have the highest propensity to progress to active, disseminated disease due to the age related deficiencies of key immunologic factors. Children aged 5-10 years have the lowest risk of progressive disease while adolescents carry higher risks, including reactivation of *M. tuberculosis* manifesting as active disease after years of successful containment.

Figure 4.1.1: Bimodal Pattern of TB Disease in Children



Complications of primary infection occur when:

- Parenchymal lesions enlarge or caseate, or nodes enlarge, compress or erode through a bronchus causing wheezing, segmental pneumonia or atelectasis.
- Occult or subclinical bacteremia seed distant sites, leading to severe forms such as miliary and CNS TB, particularly in children under age 5 years.

Children who develop primary TB usually do so within 2 years following exposure and infection, 90% by the first year. A small proportion of children with TB (usually

adolescents) develop post-primary TB either due to reactivation, after a latent period of dormant bacilli acquired from a primary infection or by re-infection.

M. tuberculosis is capable of infecting nearly any organ. However, the most common clinical manifestations of disease are found within the thoracic cavity and peripheral lymph nodes. Common intrathoracic manifestations include mediastinal or hilar lymphadenopathy and pulmonary parenchymal lesions. Less commonly, the pleura or pericardium become involved.

Extra-thoracic manifestations are common in children and make up approximately 20 - 40% of TB cases. The most commonly involved extra-thoracic sites are the peripheral lymph nodes or the central nervous system. Disease within the central nervous system represents the most serious complication of TB, with significant morbidity or mortality occurring in approximately 50% of cases.

Other extra-pulmonary sites include larynx, lymph nodes, pleura, bones, joint, skin, middle ear and mastoid. Gastrointestinal TB can mimic inflammatory bowel disease. Renal TB is unusual in younger children but can occur in adolescents.

Clinical Epidemiology Key Points

- Most cases occur in young children (<5 years).
- Most disease occurs within 2 years after exposure/infection – the majority within 1 year.
- Most cases in children are pulmonary TB – Smear negative or smear not done are the majority.
- Extra-pulmonary TB is also common and the type depends on age.
- Smear positive disease is usually in older children.

4.2 Clinical Features

Systemic symptoms and signs that are not specifically related to the organ or tissue involved include:

- Fever
- Anorexia
- Weight loss
- Failure to thrive
- Malaise
- Night sweats

Pulmonary infection is most often manifested by:

1. Chronic cough for > 2 weeks without improvement following other appropriate therapies such as broad spectrum antibiotics or treatment for malnutrition
2. Weight loss or failure to gain weight
3. Fever and/or night sweats
4. Fatigue, reduced playfulness, decreased activity

Extra-pulmonary TB is common in children and presentation varies with age as shown in Appendix II. Symptoms vary depending on site of disease, and characteristically are persistent, progressive and may be associated with weight loss or poor weight gain. Other signs that may be associated with TB in children include persistent lymphadenopathy, splenomegaly, and pallor secondary to anaemia.

4.3 WHO Case Definitions

The diagnosis of TB refers to the recognition of an active case, i.e. a patient with symptomatic disease (due to *M. tuberculosis* infection). Similarly, to adults, the type of TB disease should also be defined to enable appropriate treatment to be given and the outcome of treatment evaluated. The case definition is determined by the:

- Site of disease
- Result of any bacteriological tests
- Severity of TB
- History of previous anti-TB treatment

Tuberculosis is a Class 1 notifiable disease and all children with TB should be notified as detailed in Chapter 7.

Tuberculosis in children may be classified as smear-positive pulmonary TB, smear-negative pulmonary TB or extra-pulmonary TB; and as a new case or a previously treated case. Standard case definitions are provided below. WHO also recommends that all cases of childhood TB should be registered and reported within age bands: 0-4 years and 5-14 years.

Smear Positive Pulmonary TB

To be classified as smear positive pulmonary TB, one of the following criteria must be present:

- Two or more initial sputum smear examinations positive for acid-fast bacilli
- One sputum smear examination positive for acid-fast bacilli plus CXR abnormalities consistent with active pulmonary TB, as determined by a clinician
- One sputum smear examination positive for acid-fast bacilli plus sputum culture positive for *M. tuberculosis*

Adolescents, or children of any age with complicated intrathoracic disease, are more likely to have sputum smear-positive pulmonary TB.

Smear Negative Pulmonary TB

Many children are unable to produce sputum and are therefore classified as pulmonary TB smear negative. Diagnostic criteria for sputum smear-negative pulmonary TB should include:

- At least three sputum specimens negative for acid-fast bacilli
- Radiological abnormalities consistent with active pulmonary TB

- No response to a course of broad-spectrum antibiotics
- Decision by a clinician to treat with a full course of anti-TB chemotherapy

Extra-pulmonary TB

Children with **only** extra-pulmonary TB should be classified under this case definition. Those who have both pulmonary and extra-pulmonary TB should be classified as pulmonary TB.

Drug Resistant TB

Drug resistant TB can occur in children. Drug-resistant TB should be suspected if there is a history in the index case of:

- Exposure to a known case of drug-resistant TB
- Sputum remains smear-positive after 3 months of treatment
- History of previously treated TB
- History of treatment interruption

Features of a child suspected of having drug-resistant TB:

- Contact with a known case of drug-resistant TB
- Not responding to the anti-TB treatment regimen
- Recurrence of TB after adherence to treatment

The diagnosis and treatment of drug-resistant TB in children is complex and should be carried out at the Bustamante Hospital for Children, the paediatric referral hospital.

4.4 Diagnosis of TB Disease in Children

An accurate diagnostic test for paediatric TB does not exist. TB in children is often a clinical diagnosis and, a negative test does not rule out disease in children.

Approach to diagnosis:

1. Medical history
2. Clinical examination including growth assessment
3. Tuberculin skin test
4. Bacteriological confirmation whenever possible
5. Investigations relevant for suspected PTB or suspected EPTB
6. HIV testing

1. Medical History

The history includes social, family and medical information on the patient:

- History of exposure to a person who has infectious TB
- Symptoms of TB
- Past TB infection or disease
- Risk factors
- Contact history:
 - Closeness of contact
 - Sputum smear result of index case (if known)
 - Timing of contact, children usually develop TB within 2 years after exposure and most (90%) within the first year
 - If no index case is identified, always ask about anyone in household with cough and request assessment of that person for possible TB

2. Clinical Examination

There are no specific features on clinical examination that can confirm pulmonary TB. Common signs that should prompt an investigation into the possibility of childhood TB are shown in Table 4.4.1.

Table 4.4.1: Clinical Features of Paediatric Tuberculosis

Pulmonary	Extra-pulmonary	Systemic
Dull percussion note	Gibbus (vertebral TB)	Weight loss or failure to thrive
Crepitations	Enlarged lymph nodes	Fever
Decreased or absent breath sounds	Meningitis not responding to antibiotic treatment	Lethargy
	Pleural effusion	
	Pericardial effusion	
	Distended abdomen with ascites	
	Non-painful enlarged joint	
	Signs of tuberculin hypersensitivity (e.g. phlyctenular conjunctivitis, erythema nodosum)	

The TST can also be used as an adjunct in diagnosing TB in children with signs and symptoms of TB.

3. Tuberculin Skin Test (TST)

A positive TST occurs when a person is infected with *M. tuberculosis*, but does not necessarily indicate disease. However, the TST can also be used as an adjunct in diagnosing TB in children with signs and symptoms of TB.

A TST should be regarded as positive, whether they have received a Bacillus Calmette-Guerin (BCG) vaccination or not, under the following circumstances:

i. Induration 5 mm diameter or greater

- Children in close contact with a known or suspected contagious person with tuberculosis disease
- Children suspected to have tuberculous disease (clinical and /or radiological findings)
- Children with immunosuppressive conditions (e.g. HIV infected, on immunosuppressive therapy)

ii. Induration 10 mm or greater

- Children at increased risk of disseminated tuberculous disease (Children younger than 4 years or children with other medical conditions such as Hodgkin's disease, lymphoma, diabetes, chronic renal failure, malnutrition)

- Children with a likelihood of increased exposure to tuberculous disease (e.g. children frequently exposed to adults who are HIV infected, homeless, users of illicit drugs, residents of nursing homes, incarcerated or institutionalized)

iii. Induration 15 mm or greater

- Children age 4 years or older without any risk factors

Value of the TST in children

- The TST can be used to screen children exposed to TB (such as from household contact with TB), though children can still receive chemoprophylaxis even if the TST is not available.
- Due to the high risk of progression to severe forms of TB in the under 5 age group, the identification, screening and treatment of Latent TB Infection and active TB disease in potential adult and adolescent contacts is paramount in protecting this vulnerable group.
- The TST is useful in HIV-infected children to identify those with dual TB/HIV although immunosuppressed children may have a negative TST.
- A negative test does not exclude disease and a positive test does not differentiate latent TB Infection from active TB disease.

4. Bacteriological Confirmation

A bacteriological specimen must be collected on initial examination. Bacteriological confirmation is ideal.

Common ways of obtaining samples for smear microscopy include the following:

a. Expectoration

In children under 5 years of age, sputum is difficult to obtain. However, in children who are able to produce a specimen, it should be collected for bacteriological confirmation. Older children (10 years of age or older) are usually able to provide a sputum sample. Three sputum specimens should be obtained:

- An on-the-spot specimen (at first evaluation)
- An early morning specimen
- A second on-the-spot specimen (at a follow-up visit).

The procedure for sputum specimen collection, handling and transfer is detailed in Chapter 8.

b. Gastric aspiration

Gastric aspiration using a nasogastric feeding tube can be performed in young children who are unable to expectorate sputum. A gastric aspirate should be obtained on three consecutive mornings.

- Patients should have nil per os (NPO) for eight hours prior to the procedure.
- Aspirate the stomach contents first. Then instil no more than 50 ml of sterile distilled water. Aspirate the contents of the stomach and add the aspirate as the first specimen.
- The specimen should be delivered to the laboratory within 4 hours of collection because acid is detrimental to mycobacteria. If that is not possible, it should be directly placed into a buffered solution (refer to Chapter 8).

c. Sputum induction

Sputum induction is safe and effective in children of all ages and the bacterial yields are as good as or better than for gastric aspirates. However, training and specialized equipment are required to perform this procedure properly. Patients should be kept NPO for eight hours before the procedure.

The sputum induction procedure can be administered by a physiotherapist using 4 steps:

- Administration of salbutamol.
- Delivery of hypertonic saline by nebulizer.
- Chest percussion, vibration, or active breathing performed by a trained technician.
- Sample collection following spontaneous expectoration or suction of the naso-oropharynx should be obtained on each of three consecutive mornings.

5. Chest X-ray

A chest x-ray is an important tool for diagnosing PTB in children. The most common radiological feature is hilar or para-tracheal lymphadenopathy which tends to be asymmetrical. **Frontal and lateral chest** radiographs are required to detect this.

Parenchymal lesions can be anywhere in primary disease. Typically, in adolescents the upper lobes are involved. Infants can present as severe pneumonia with extensive parenchymal disease and respiratory distress that is challenging to

differentiate from the many other possible causes of pneumonia in infants. Note that a contact history is very important and often positive in infants with TB disease.

Cavitation is rare but may be present in very young or immunocompromised children with adult type disease or with caseating pneumonia secondary to lymphobronchial disease.

An experienced radiologist should ideally review films.

6. HIV testing

As in all cases of TB, an HIV test is mandatory in the assessment of children with TB.

Administering treatment and ensuring adherence

Children, their parents and other family members, as well as other caregivers, should be educated about TB and the importance of adherence and completing treatment.

The support of the child's parents and immediate family is vital to ensuring a satisfactory outcome of treatment. Often a health-care worker can observe or administer treatment, but if this arrangement is not convenient for the family, a trained community member (preferably someone other than the child's parent or immediate family) can undertake this responsibility. All children should receive treatment free of charge, whether the child is smear-positive at diagnosis or not. Fixed-dose combinations of drugs should be used whenever possible to improve simplicity and adherence. Patient treatment cards are recommended for documenting treatment adherence.

Specialist management is warranted in cases with:

- Severe forms of PTB and EPTB (such as TB meningitis and miliary TB) for further investigation and initial management
- Severe malnutrition for nutritional rehabilitation
- Signs of severe pneumonia (i.e. chest in-drawing) or respiratory distress
- Other co-morbidities e.g. severe anaemia
- Severe adverse events, such as clinical signs of hepatotoxicity (e.g. jaundice)
- Diagnostic uncertainty requiring further investigation at referral level
- Necessary for HIV-related care e.g. to commence ART

Children require close clinical monitoring, which should include symptom assessment, weight measurement, assessment of adherence to treatment and enquiry about any

adverse events. Dosages of anti-TB drugs should be adjusted to account for any weight gain.

For the continuation phase, if it is not possible to ensure good adherence and treatment outcome on an outpatient basis, some children may require hospitalization for social or logistical reasons. Chest X-ray (CXR) 2 months into treatment should be done to rule out extension of disease. Persistent radiographic signs are not an indication to change treatment if there is clinical improvement. Residual scarring may persist for 2-3 years. A normal CXR is not necessary to discontinue treatment. Patients should be followed for 1 year after treatment completion to monitor clinical health and resolution of radiographic findings.

4.5 Standardized Treatment Regimens

The recommended treatment regimens and dosages for the treatment of childhood TB are similar to adults and are depicted in Table 4.5.1

Table 4.5.1: Anti- Tuberculosis Drugs and Dosages

Drugs	Dosage Formulation	Daily Dosage (Range) mg/kg	Maximum Dose
Ethambutol	Tablets 100mg 400mg	20 (15-25)	2.5g
Isoniazid	Scored Tablets 100mg 300mg	10 (7 -15)*	Daily 300mg
Pyrazinamide	Scored tablets 500mg	35 (30-40)	2g
Rifampin	Capsules 150mg 300mg Syrup formulated capsules	15 (10-20)	Daily 600mg
Pyridoxine (has no Anti-TB activity. Used to prevent Isoniazid neuropathy)	25 mg tablet 50 mg tablet	1mg/kg	25 mg

Dosages are reviewed at each visit and adjusted according to weight. As children approach a body weight of 25 kg, clinicians can use adult dosing recommendations.

Table 4.5.2 shows the anti-TB regimens by anatomical site and drug resistance.

Table 4.5.2: Anti-TB Drug Regimens by Anatomical Site and Drug Resistance

Anti-TB Drug Regimens		
	Intensive Phase	Continuation Phase
Pulmonary and extra-pulmonary (except meningitis and osteoarticular TB)	2 months RIPE	4 months IR
Tuberculous meningitis and osteoarticular TB	2 months RIPE	10 months IR
MDR-TB	Individualized Regimes	Individualized Regimes

Smear-positive cases should follow the standard NTP follow-up microscopy.

Therapy should be extended from 4 to 7 months for a total of 9 months of therapy if:

- Positive sputum culture after 2 months of treatment.
- Having a cavity on chest x-ray at the end of treatment.
- Having more extensive disease and/or cavities on a chest x-ray in the first 2 months of therapy.

HIV infected persons receive the same duration as HIV negative patients once ART is commenced as per treatment guidelines. Treatment should also be extended if there is treatment interruption.

Treatment Response

Most children with TB, including those who are HIV-infected, have a good response to the 6-month regimen. Treatment response is commonly noted by the end of the intensive phase and is indicated by resolution of symptoms and weight gain.

If there is a poor response to TB treatment, review:

- Adherence
- Alternate diagnosis to TB
- Drug-resistance
- Dosing regimen
- Failure to treat co-morbidity such as HIV

4.6 Special Populations

- HIV infected children
- Adolescents
- The exposed child
- The exposed newborn

HIV Infected Children

HIV-infected children are at 20-times greater risk of TB disease than HIV-uninfected children and at much higher risk of TB-related death. Those who are infected with HIV have impaired cell-mediated immunity to control TB infection, conferring a higher risk of progression to active TB disease after exposure; of reactivation of latent infection; and of severe disease manifestations. Timely recognition of HIV and initiation of antiretroviral therapy (ART) is essential for immune restoration and improved TB control.

It is recommended that HIV-infected children receive:

1. Four first-line drugs (RIPE) in the intensive phase for suspected or confirmed drug-sensitive TB irrespective of severity of disease
2. Similar duration of regimens as for HIV-uninfected
3. ART as recommended within 2-8 weeks of starting TB treatment or continue ART
4. Co-trimoxazole preventive therapy
5. Pyridoxine supplement
6. Nutritional support

Where possible, HIV-infected children should be treated with rifampin for the entire treatment duration, as higher relapse rates among HIV infected adults have been found when ethambutol is used in the continuation phase.

There has been caution in the use of ethambutol in children too young to report early visual deterioration, but ethambutol has been safely used in infants and young children at recommended dosages. Children on ethambutol should receive visual assessment before and after treatment.

ART regimens for children in need of TB treatment

All newly diagnosed TB cases that are also HIV positive and are ARV naïve should be started on TB therapy immediately and initiate ARV treatment within 2 to 8 weeks. HIV positive TB patients with profound immunosuppression (e.g. CD4 counts less than 50

cells/ mm³) should receive ART within the first two weeks of initiating TB treatment. Table 4.6.1 shows the recommended regimen for children and infants starting ART while on TB treatment.

Table 4.6.1 Recommended Regimen for Children and Infants Starting ART While on TB Treatment

Age	Regimen
Younger than 3 years	Triple NRTI (AZT + 3TC + ABC)
3 years and older	2 NRTIs + EFV or Triple NRTI (AZT + 3TC + ABC)

Ensure optimized dosing of rifampin based on new dosing guidelines. Triple Nucleoside Reverse Transcriptase Inhibitors (NRTI) is recommended only for the duration of TB treatment; an age-appropriate Protease Inhibitor (PI) or Non-nucleoside reverse transcriptase inhibitors (NNRTI)-based regimen should be restarted or switched at termination of rifampin-based therapy.

Table 4.6.2 shows the recommended regimen for children and infants already on ART starting TB treatment.

Table 4.6.2: Recommended Regimen for Children and Infants Already on ART who are Starting TB Treatment

Child	Age	Regimen
Child on ART regimen containing NVP or LPV/r	Younger than 3 years	Triple NRTI (AZT + 3TC + ABC)
Child on standard NNRTI-based regimen (2 NRTIs + EFV or NVP)	3 years and older	If on 2 NRTIs + EFV, continue this regimen or If on 2 NRTIs + NVP, change NVP to EFV or Triple NRTI (AZT + 3TC + ABC)
Child on standard PI-based regimen (2 NRTIs+ PI)	Younger than 3 years	Triple NRTI (AZT + 3TC + ABC) or Continue LPV/r and consider adding RTV to achieve full therapeutic dose
Child on standard PI-based regimen (2 NRTIs+ PI)	3 years and older	If no history of NNRTI-based regimen failure: Change LPV/r to EFV

		or Change entire regimen to Triple NRTI (AZT + 3TC+ ABC) or If triple NRTI is contraindicated, continue LPV/r and consider adding RTV to achieve full therapeutic dose. If history of NNRTI-based regimen failure: Triple NRTI (AZT + 3TC+ ABC) or Continue LPV/r and consider adding RTV to achieve full therapeutic dose. Consider consultation with experts for construction of second-line regimen.
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Adolescents

Adolescence is a uniquely susceptible time for disease progression. Disease presentation can vary from childhood manifestations to adult type TB, with cavitary pulmonary TB or extra-thoracic manifestations with longer incubation periods such as genitourinary TB.

Health care workers should inquire about substance abuse, anxiety or depression which may be exacerbated by infection control measures or contact investigation procedures and stigma associated with the disease. Adherence and follow up may be affected by these factors compounded by behavioural aspects of adolescence resulting in poor outcomes.

The TB Exposed Child and Management of Contacts

A child with tuberculosis represents a sentinel event of ongoing transmission in the community. Prompt evaluation and treatment of children exposed to an infectious adult source is the most efficient way to prevent pulmonary TB.

- All exposed children should have a symptom enquiry
- Children less than 5 years and symptomatic children should have a physical examination and chest x-ray

Window therapy

- All children less than 5 years with a negative TST and no active disease by examination or radiology should be given “window” preventative therapy to prevent the development of TB
- This is because it takes up to 8 weeks after infection for the TST to convert to positive during which time the infection will progress to disease
- Isoniazid is used for presumed drug susceptible exposure
- Isoniazid may be discontinued if after 8 weeks, after the last contact, the TST is negative, the child remains asymptomatic, is immunocompetent and is greater than 6 months old
- For children < 6 months please see management of the newborn exposed to TB

In the exposed child, if the TST is positive and there is no clinical or radiographic disease, then a full course of treatment for LTBI is recommended. When a new active child is the index case, vigorous reverse contact tracing should be carried out to find the source case. Most source cases are household contacts but non household contacts such as caregivers should be sought. For patients admitted to hospital, adolescent or adult visitors should be screened for active TB.

Congenital Tuberculosis

This is a relatively uncommon event but a high index of suspicion must be maintained. Spread occurs via the haematogenous route direct from reproductive organs or postnatally from respiratory transmission. Symptoms are nonspecific and indistinguishable from bacterial sepsis or congenital viral infections and manifest as early as 2-3 weeks of age. Infants born to mothers with active or confirmed TB or LTBI should be managed according to the scenarios in Table 4.6.3.

Evaluation of congenital TB includes clinical examination, TST, cultures including lumbar puncture, chest x-ray, abdominal ultrasound and cranial ultrasound. The initial TST will be negative but becomes positive 1-3 months after treatment.

Table 4.6.3: Management of the Newborn Infant Exposed to TB

Scenario	Treatment of Mother (or household contact)	Treatment of Infant
1. Mother (or household contact) is asymptomatic and has a positive TST	• Manage mom as LTBI and evaluate other household members for possible contagious contact • Allow breastfeeding	• No special evaluation or therapy • No separation from mother • Give BCG vaccine

result and normal CXR		
2. Mother or household contact has clinical signs and symptoms or abnormal findings on chest radiograph consistent with TB disease	- Investigate all household members - Separate the infant from mother (or contact) until mother (or contact) and infant are receiving appropriate anti TB therapy and mom is non-infectious, adherent and clinically improving - Mother with drug susceptible TB may breastfeed after 2 or more weeks after initiating therapy and is not considered contagious	- Exclude congenital TB - Once excluded isoniazid is given until the infant is 3 or 4 months of age - Once isoniazid is commenced separation is not necessary unless mom has MDR TB - TST should be performed at 3-4 months, if positive the infant should be reassessed for TB disease - If TB disease is excluded the infant with a positive TST should receive isoniazid for 9 months - Evaluate infant monthly for signs of illness or poor growth - If TST result is negative at 3-4 months and mother (or contact) has good treatment response, is adherent and no longer contagious isoniazid should be discontinued - After completion of isoniazid give BCG vaccine
3. Mother (or household contact) has positive TST, and abnormal chest X-ray and no evidence of active TB disease	- If CXR considered to be an old healed TB scar and mom not previously treated, she should have a sputum test - Treat mother for LTBI - Evaluate household contacts - No separation required	- Infant considered low risk but should be evaluated clinically and radiologically at birth and followed up with mother. - Allow breastfeeding - Give BCG vaccine
4. Mother treated for TB	Mother should have follow up smear exam to confirm she is no longer infectious	Evaluate infant for congenital TB and treat if confirmed.

during pregnancy	• Examine placenta histologically for AFB and granulomata and culture for MTB • Separation is not indicated if treatment is adequate and mom is clinically improving and no longer infectious • Mother with drug susceptible TB may breastfeed after 2 or more weeks after initiating therapy and is not considered contagious	• If congenital TB is excluded and mother is confirmed non-infectious and no other household contacts have TB disease isoniazid is not necessary and BCG can be given • If in doubt refer to scenario 2
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4.7 Breast Feeding Women

Once a pregnant woman has been on treatment for at least 2-3 weeks, she is generally no longer infectious. If a pregnant woman with TB has been on treatment for TB for several weeks before delivery, it is less likely that the baby will become infected. The risk is highest, if a mother is diagnosed at the time of delivery or shortly thereafter. If a pregnant woman is found to have pulmonary TB shortly before delivery, then the baby, and possibly the placenta should be investigated for evidence of congenital TB infection, and if found, the baby treated.

4.8 BCG Vaccination

The HIV pandemic has implications for BCG vaccination. Although there have been a few reports of disseminated BCG disease after BCG immunisation of HIV-infected children, prospective studies comparing BCG immunisation in HIV-infected and uninfected infants have shown no difference in risk of complications. It is recommended that BCG vaccination policy should depend on the prevalence of TB in a country. In countries with a high TB prevalence, the benefits of BCG vaccination outweigh the risks, and WHO recommends a policy of routine BCG immunisation for all neonates.

BCG vaccine should not be given to symptomatic HIV infected children because of the risks of disseminated BCG disease. In Jamaica, the National Expanded Programme on Immunisations recommends routine BCG immunisation in the neonatal period to all infants.

Management of BCG disease in HIV-infected children (or children with other immuno-deficiencies)

The diagnosis of BCG disease is difficult and the treatment is specialized, as *M. bovis* is resistant to pyrazinamide and requires higher doses of other first-line TB medications – e.g. some experts recommend a daily dose of isoniazid of up to 15 mg/kg (maximum 300mg) and a daily dose of Rifampin of up to 20 mg/kg (maximum 600 mg). HIV-infected children suspected of having BCG disease should be referred to an appropriate expert for management.

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CHAPTER: V

HIV/TUBERCULOSIS CO-INFECTION

CHAPTER: V

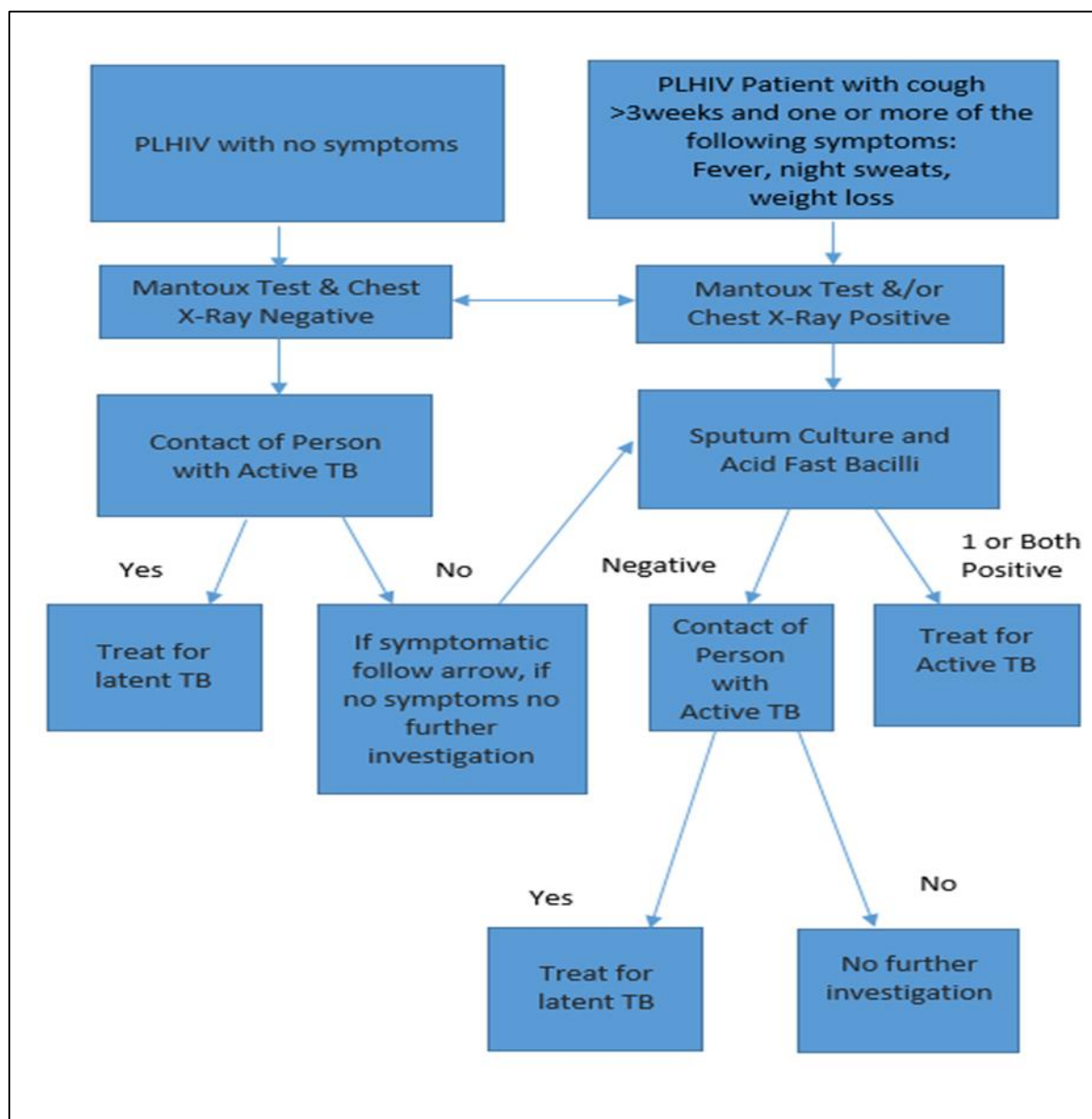
HIV/TUBERCULOSIS CO-INFECTION

5.1 Management of TB/HIV Co-Infection

5.1 HIV/Tuberculosis Co-infection

Persons infected with HIV have a greater risk of infection with Tuberculosis and developing active TB. Figure 5.1.1 outlines the clinical algorithm for the diagnosis and management of HIV/Tuberculosis co-infection. All persons infected with HIV must be screened and if indicated tested for TB; and if found to have either latent or active TB must be effectively treated. Conversely, all persons with TB must be tested for HIV and appropriately managed.

Figure 5.1.1: Clinical Algorithm for the Investigation and Diagnosis of HIV/Tuberculosis Co-infection



TB and HIV/AIDS Standards of Treatment and Management

1. Prior to initiating any TB therapy, review all medications that the individual is taking and assess for potential drug interactions with TB medications.
2. In TB/HIV co-infection, initiation of treatment for TB takes precedence over the treatment of HIV and should be commenced first. Persons already on HAART can be continued with appropriate modifications, both in anti-retroviral therapy (ART) and anti TB treatment.
3. In persons who were not previously on ART, the need and timing of ART initiation is decided after assessing the risk of HIV disease progression.
4. Concurrent administration of rifampin with certain HIV medications (ART) may require dose adjustments or switches to alternative ART or anti-tuberculous drugs to reduce both interactions and toxicity.
5. A CD4 count should be obtained.
6. All patients being treated for TB should receive a daily DOT regimen for the entire course of therapy. **Do not switch to a two times weekly or three times weekly regimen.** (PAHO, 2017)
7. TB drugs should be calculated according to mg/kg body weight. This should be done on the lower figure in the range and rounded up to the next available dose supplied by the manufacturer (***For dosage table, see Chapter 3.***)
8. HIV-positive individuals treated with isoniazid should receive Pyridoxine (B₆) 25 mg daily on the same schedule as isoniazid.
9. Ingestion of 9 months of isoniazid or 4 months of rifampin is considered adequate preventive therapy.
10. All HIV/TB patients need to be monitored closely for response to treatment and signs of treatment failure or relapse; if failure or relapse is suspected, assume rifampin resistance until culture-confirmed otherwise. Treatment modification will require consultation with a physician who is an expert in the management of TB.

Monitoring

- a. Pre-treatment evaluation
 - Do a chest x-ray, symptom review and complete physical examination to exclude active tuberculosis disease. If active disease is excluded, no further chest x-rays are recommended for the patient who remains asymptomatic.
 - Obtain medical history including:
 - Previous adverse reaction to any TB drugs
 - Underlying liver disease
 - Previous Isoniazid associated liver injury

- Current medications
- Obtain a baseline liver profile and viral hepatitis screen and manage abnormalities as appropriate.
- Obtain a baseline CBC including platelets and appropriately manage any abnormalities detected.

b. Monitoring during treatment

Clinical assessment should be performed on at least a monthly basis for all patients being treated for TB. Patients should be assessed for:

- Adherence to the prescribed regimen
- Signs and symptoms of adverse drug reactions
 - If individual is on rifampin - at each visit evaluate for potential interactions with HIV drugs or any other medication.
 - Monthly lab monitoring, done at baseline and at the end of the first two months of therapy, thereafter symptom driven or as indicated.
 - Manage individuals with asymptomatic liver enzyme elevations, adverse reactions or hepatotoxicity.

Tuberculosis Disease in Patients with HIV/AIDS

The treatment regimen in HIV/AIDS patients may involve different TB drugs, daily dosing and may require a longer treatment duration.

1. Drugs used to treat TB in HIV/AIDS adults ≥ 15 yrs.
 - a. If the patient is on antiretroviral therapy, the patient's TB treatment regimen should be reviewed with a specialist physician.
 - b. Individuals should receive the usual 4-drug TB regimen (Rifampin, Isoniazid, Pyrazinamide and Ethambutol) if they are not on antiretroviral therapy or on NRTI-class drugs (e.g. Zidovudine) and NNRTIs (e.g. Efavirenz or Nevirapine).
2. Individuals receiving any other antiretroviral will need to have Rifabutin (RBT) substituted for Rifampin; this should be done in consultation with a physician knowledgeable about treating HIV/AIDS and TB.
 - a. Monitoring of patients taking Rifabutin should include complete blood count looking for neutropenia and thrombocytopenia; the patient should also be asked about uveitis (eye pain, visual changes).
 - b. Other drugs, for example clarithromycin and fluconazole can increase RBT levels and cause RBT toxicity.

3. When switching from a rifampin based regimen to a RBT or non-rifampin based regimen, wait at least 2 weeks before initiating antiretroviral drugs to prevent the interference of rifampin with the antiretroviral drug.
4. Individuals who are unable to take a Rifampin or Rifabutin-based regimen, should be managed as follows using DOTs:
 - a. Individuals on an initial regimen of Isoniazid, Rifampin/Rifabutin, Pyrazinamide and Ethambutol:
 - Discontinue Rifampin/Rifabutin; discontinue Pyrazinamide after 2 months and continue to treat with Isoniazid and Ethambutol for a total of 18 months.
OR
 - Discontinue Rifampin; add Streptomycin and continue to treat with Isoniazid, Pyrazinamide, Ethambutol and Streptomycin for 2 months. Discontinue Ethambutol after 2 months and continue to treat with Isoniazid, Pyrazinamide and Streptomycin for 7 months.
 - If Streptomycin cannot be given for the total duration of treatment or for at least 4 months following culture conversion, add Ethambutol back into the regimen and treat for a total of 12 months.
5. Therapeutic drug monitoring is suggested for patients with risk factors for altered drug absorption or metabolism and excretion. This includes:
 - a. Gastrointestinal disease or HIV infection, due to the potential of malabsorption of drugs.
 - b. Liver or renal disease, due to reduced drug excretion.
 - c. Drug resistant TB disease, to allow optimal availability of all drugs
6. Duration of treatment
 - a. The minimum duration of short-course rifampin or Rifabutin-containing TB regimens is 26 weeks.
 - b. If Pyrazinamide is not started within the first 2 weeks of treatment and continued for the first 8 weeks of therapy, or if Pyrazinamide is contraindicated, a minimum of nine months of Isoniazid and Rifampin is required and the first 8 weeks of treatment must be administered daily by DOT.
 - c. If at the end of 8 weeks of therapy, the individual does not experience resolution of symptoms of TB, or if their sputum culture does not convert to negative, treat for a total of 9 months, to include a minimum of 4 months after documented culture conversion.

7. Patients with HIV/TB coinfection should also receive co-trimoxazole as prophylaxis. Consideration for continuation after TB treatment is completed is in accordance with HIV national treatment guidelines.
8. Regimens for Pregnant Women
 - a. The only difference between treating HIV-positive and HIV-negative pregnant women is the use of Pyrazinamide. The benefit of optimal treatment of TB in HIV-positive pregnant women outweighs the very small risk of using Pyrazinamide in pregnancy. Therefore, for HIV/AIDS and TB in pregnant women, follow the treatment recommendations above.
 - b. Streptomycin is contraindicated for pregnant women
9. Regimens for Infants and Children <15 yr.
 - a. Standard TB therapy of Isoniazid, Rifampin, Pyrazinamide, and Ethambutol with daily DOT is given for eight weeks.
 - Discontinue Ethambutol when initial culture results confirm the organisms are fully susceptible to Isoniazid and Rifampin.
 - Discontinue Pyrazinamide after 8 weeks.
 - If susceptibility results are not done or unknown, give Isoniazid, Rifampin, Pyrazinamide, and Ethambutol followed by Isoniazid and Rifampin for 18 weeks.
 - Individuals too young to have visual acuity or red-green perception testing should receive Ethambutol at a dosage of 15-mg/kg body weight, or Ethambutol may be omitted at the discretion of the prescribing physician.
 - b. Due to the interaction with Rifampin, individuals receiving antiretroviral therapy should have a consultation with a HIV/TB clinical specialist.
 - c. Infants and children with meningeal, bone, joint or miliary TB should receive a minimum of 12 months of therapy.
10. Regimens for Extra-pulmonary TB
 - a. Individuals with TB at any site should have a chest x-ray and if able to produce sputum, a sputum specimen for smear or culture.
 - b. Adults with meningeal, bone and joint TB should be treated for a minimum of 9 months, and may be extended to 12 months total. All other sites should be treated with the same drug regimens recommended for pulmonary TB.
11. Regimens for Isoniazid Resistance or Intolerance
 - a. Individuals on initial regimen of Isoniazid, Rifampin, Pyrazinamide and Ethambutol:
 - Discontinue Isoniazid; continue to treat with Rifampin, Pyrazinamide and Ethambutol for a total of 6 months or 4 months after documented culture conversion, whichever is longer.

OR

- Discontinue Isoniazid and Pyrazinamide; continue to treat with Rifampin and Ethambutol for minimum of 12 months.
- b. Individuals on an initial regimen of Isoniazid, Rifampin and Ethambutol:
 - Discontinue Isoniazid; continue to treat with Rifampin and Ethambutol for a minimum of 12 months.

12. Regimens for Rifampin Resistance or Intolerance

- a. Individuals on initial regimen of Isoniazid, Rifampin, Pyrazinamide and Ethambutol:
 - Discontinue Rifampin; discontinue Pyrazinamide after 2 months and continue to treat with Isoniazid and Ethambutol for a total of 18 months.

OR

- Discontinue Rifampin; add Streptomycin and continue to treat with Isoniazid, Pyrazinamide, Ethambutol and Streptomycin for 2 months. Discontinue Ethambutol after 2 months and continue to treat with Isoniazid, Streptomycin and Pyrazinamide for 7 months.
 - If Streptomycin cannot be given for the total duration of treatment or for at least 4 months following culture conversion, add Ethambutol back into the regimen and treat for a total of 12 months.

13. Multi-Drug Resistant TB (MDR-TB)

- MDR-TB is TB resistant to both Isoniazid and Rifampin.
- Treatment must be individualized and prolonged based on medication history and susceptibility studies.
- Give at least 3 drugs to which the organism is susceptible. The regimen should continue until sputum conversion is documented, followed by at least 12 months of a two-drug therapy.
- Consult the TB Consultant.

14. Monitoring is the same as for HIV-negative TB patients.

Immune reconstitution inflammatory syndrome

Immune reconstitution inflammatory syndrome (IRIS) is an inflammatory reaction that can occur when the immune system of an HIV-positive patient improves with anti-retroviral therapy. In TB/HIV co-infected patients, IRIS may occur in up to 33% of patients and usually occurs within the first three months of starting therapy. IRIS associated with TB may present with fever, appearance of new adenopathies, increased size of pre-existing adenopathy, pain or respiratory symptoms.

Management of IRIS includes:

- Treatment of opportunistic infection to reduce the antigenic load
- Continuation of ART
- Use of anti-inflammatory drugs or corticosteroids

Severe reactions may occur which include respiratory failure or tracheal compression due to massive adenopathy. These cases require support and corticosteroid therapy (prednisone 0.5mg/kg/day for five to ten days). (PAHO, 2017)

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CHAPTER: VI

LATENT TUBERCULOSIS INFECTION

CHAPTER: VI

LATENT TUBERCULOSIS INFECTION

6.1 Latent Tuberculosis

6.2 Who to Test

6.3 Diagnosis of Latent Tuberculosis

6.4 Treatment of Latent Tuberculosis

6.1 Latent Tuberculosis

Latent Tuberculosis infection (LTBI) is a state of persistent immune response to stimulation by *Mycobacterium tuberculosis* antigens, with no clinically manifested evidence of active TB disease. Up to one third of the world's population is estimated to be infected with *M. tuberculosis*, the majority of which have no signs or symptoms of active TB disease, and although not having active TB disease currently, are still at risk for active TB disease and becoming infectious. The risk of developing active TB disease is approximately 5-10% (WHO, 2015) and usually occurs within the first five years of initial infection. This is dependent on several factors, significantly immunological state.

Population wide testing for LTBI is not feasible and not recommended due not only to the cost but also due to unproven public health impact and the risk for drug-related adverse side effects. However, in certain population groups where the risk of progression to active disease is higher, the benefits of treatment outweigh the harm.

Management of LTBI involves the following:

1. Identifying and testing those who should be tested
2. Complete safe treatment with minimal or no side effects
3. Monitoring and evaluation of the process

6.2 Who to Test

Preventive treatment of *M. tuberculosis* infection should be selectively targeted to the population groups at highest risk for progression to active TB disease and would consequently benefit most from treatment of LTBI.

Jamaica is classified as an upper middle-income country, as such the following are recommended for at-risk populations for LTBI testing and treatment:

1. *Systematic testing and treatment should be performed in:*

- Persons living with HIV - as per recommendation in Chapter 5
- Contacts of pulmonary TB - both children and adults
- Patients initiating anti-tumor necrosis factor (TNF) treatment
- Patients receiving dialysis
- Patients preparing for organ or haematologic transplantation
- Patients with silicosis

2. *Systematic testing and treatment should be considered for:*

- Prisoners
- Health Care workers
- Immigrants from high burden TB countries
- Homeless persons
- Illicit drug users

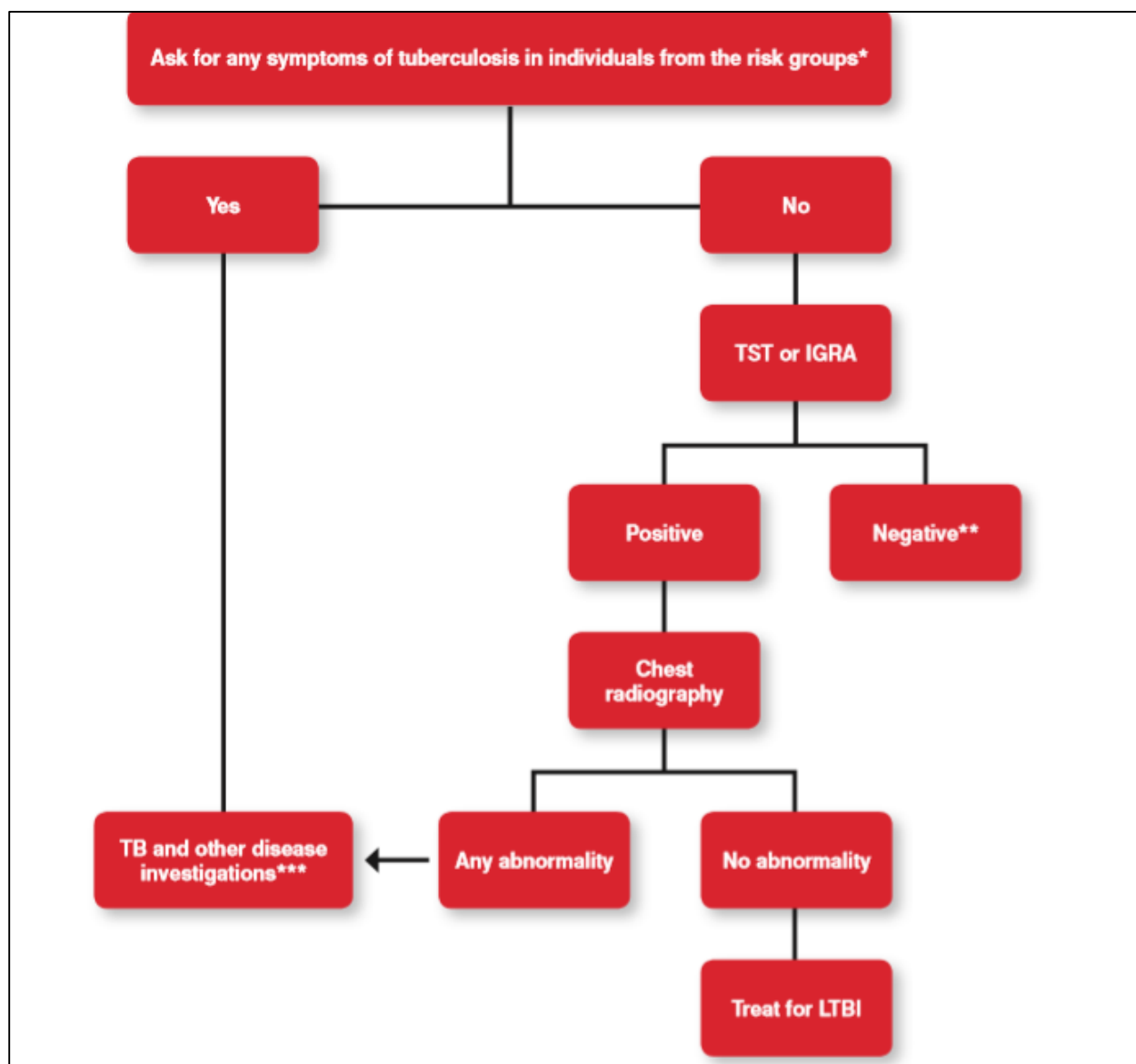
Systematic testing for LTBI is **not** recommended in people with diabetes, people with harmful alcohol use, tobacco smokers and underweight persons unless they are already included in the above recommendations.

6.3 Diagnosis of Latent TB

1. Check for active disease.
The clinical algorithm for the investigation and diagnosis of HIV/TB Co-infection (Figure 5.1) should be utilized. However, patients with latent TB are asymptomatic. For active disease patients might not have classical symptoms, and radiographs may not be typical of expected findings; hence clinicians must have a high index of suspicion when dealing with HIV positive patients.
2. Perform tuberculin skin test.
 - a. Indurations of $\geq 5\text{mm}$ are considered positive. Assess further for active versus latent disease using chest radiographs.
 - b. Indurations of $<5\text{mm}$ do not rule out the presence of disease. Correlate TST result with clinical presentation.
 - c. For persons whose initial TST was negative and the CD4 count $< 200\text{mm}^3$, the TST should be repeated once the CD4 count becomes greater than 200mm^3 .
3. If the patient had a positive TST, perform a chest X-ray to identify changes associated with TB disease:
 - HIV/AIDS individuals ≥ 5 years of age should receive a posterior-anterior chest x-ray.
 - HIV/AIDS individuals < 5 years of age should receive a posterior-anterior and lateral chest x-ray.
4. If no abnormalities are detected on the chest x-ray, the patient should be treated for LTBI.
5. If any abnormality is detected on the chest x-ray, further investigations should be done for TB and or other diseases as indicated.

Figure 6.3.1 outlines the algorithm for the targeted diagnosis and treatment of LTBI in individuals from risk groups.

Figure 6.3.1: Algorithm for Targeted Diagnosis and Treatment of LTBI in Individuals from Risk Groups (WHO, 2015)



*Symptoms of TB include any one of: cough, haemoptysis, fever, night sweats, weight loss, chest pain, shortness of breath and fatigue.

** Clients for whom LTBI treatment is not indicated should be provided with information about TB including the importance of seeking care if symptoms of TB develop.

***Follow National TB Guidelines for TB investigation. Additionally, those in whom TB is excluded after investigations (including individuals with fibrotic radiologic lesions) can be considered for LTBI treatment.

6.4 Treatment of Latent Tuberculosis

Six months daily monotherapy with Isoniazid is the treatment for both adults and children. The recommended treatment option for LTBI is detailed in Table 6.4.1 below (WHO, 2018).

Table 6.4.1: Six Month Daily Isoniazid Treatment Regimens for LTBI

	Drug Regimen	Dose per kg body weight	Maximum dose
Adults (Daily for Six months)	Isoniazid	5 mg/kg	300 mg
Children (Daily for Six months)	Isoniazid	10 mg/kg (range 7 -15mg)	300 mg

Alternate regimens to the six months of monotherapy with Isoniazid are detailed in Table 6.4.2 below (WHO, 2018).

Table 6.4.2: Alternate Treatment Regimens for LTBI

Drug regimen	Dose per kg body weight	Maximum dose
Isoniazid (daily for 6 or 9 months)	Adults – 5mg Children – 10mg (range 7 – 15mg)	300mg
Daily Rifampin alone for 3 – 4 months	Adults – 10mg Children – 15mg (range 10-20mg)	600mg
Daily Isoniazid plus Rifampin for 3 -4 months	Isoniazid Adults – 5mg Children – 10mg (range 7 – 15mg) Rifampin Adults – 10mg Children – 15mg (range 10-20mg)	Isoniazid – 300mg Rifampin – 600mg

The patient does not require admission as they are not infectious for TB. They do require follow up in light of adherence at the primary health care setting on a weekly basis. The

follow up can be done by a Community Health Aide who reports to the Public Health Nurse on the compliance of patient and completion of medication.

Isoniazid and Rifampin are both tolerated and recommended for treatment of LTBI in pregnant women and during breast feeding.

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CHAPTER: VII

TUBERCULOSIS SURVEILLANCE

CHAPTER: VII

TUBERCULOSIS SURVEILLANCE

7.1 Goal and Objectives of Tuberculosis Surveillance

7.2 Case Detection

7.3 Case Definitions

7.4 Case Investigation

7.5 Data Management

7.1 Goal and Objectives of Tuberculosis Surveillance

Tuberculosis surveillance aims to detect and measure the number of TB cases and related deaths that occur each year and to assess trends in these parameters over time.

Objectives of TB surveillance

1. Describe the number of new TB cases by category
 - Multi-Drug-Resistant Tuberculosis
 - HIV positive (co-infected)
 - Age/Sex distribution
 - Geographic distribution
2. Measure the number of TB deaths
3. Track new cases and deaths over time
4. Measure treatment outcome by:
 - Age/Sex
 - Classification (pulmonary versus extra-pulmonary)
 - Time (trends)
 - Geographic distribution

7.2 Case Detection

Medical Practitioners shall notify as Tuberculosis all patients presenting with a chronic cough (> 3 weeks duration). Tuberculosis is a Class 1 notifiable disease, therefore all suspected cases should be reported within 24 hours to the Parish Health Department or National Surveillance Unit [Public Health (Class 1 Notifiable Diseases) Order 2003, Government of Jamaica, 2003].

7.3 Case Definitions (WHO, 2013)

- a) **Suspected case** – A person who presents with a persistent cough for greater than 3 weeks.
- b) **Confirmed case**
 - Bacteriologically confirmed - Person from whom a biological specimen is positive by smear microscopy, culture or PCR (Such as Xpert MTB/RIF).
 - Clinically diagnosed case – An individual who does not fulfil the criteria for bacteriological confirmation but has been diagnosed with active TB by a clinician or other medical practitioner who has decided to give the patient a full course of TB treatment.
- c) **Pulmonary TB** - A bacteriologically confirmed or clinically diagnosed case of TB, with disease involving the lung parenchyma or the trachea-bronchial tree. Tuberculous intra-thoracic lymphadenopathy or tuberculous pleural effusion without radiographic evidence of lung lesions is classified as extra-pulmonary TB. A case with pulmonary and extra-pulmonary TB should be classified as a case of pulmonary TB. Miliary TB is classified as pulmonary TB.
- d) **Extra-pulmonary TB** - A bacteriologically confirmed or clinically diagnosed case of TB with disease involving organs other than the lungs and no pulmonary involvement (see pulmonary TB above).
- e) **New patient** - A patient with bacteriologically confirmed or clinically diagnosed Tuberculosis who has never been treated for TB or who has taken anti-TB drugs for less than one month.
- f) **A relapsed patient** - A patient who has previously been treated for TB and declared cured or treatment completed and is now diagnosed with TB (Bacteriologically Confirmed or Clinically Diagnosed).

Notification of TB Cases by Health Care Workers

All suspected cases of Tuberculosis shall be notified using a Class 1 notification form (Appendix III).

The notification form must be completed with all available information about the suspected case and transmitted to the Parish Health Department or the National Surveillance Unit within 24 hours of suspicion. The Parish Health Department must

forward notifications to the Regional Health Authority and the National Surveillance Unit as detailed in the National Surveillance Manual for Jamaica. Hospital Active Surveillance Officers are responsible for searching the wards and laboratory of each hospital for cases of tuberculosis that were not reported as above.

7.4 Case Investigation

The responsibility for case investigation lies with the Parish Medical Officer (Health) who will delegate as appropriate. Contact investigation should be carried out for all confirmed cases of TB (bacteriologically and clinically confirmed). The schedule for completing the Case Investigation Form (Appendix I) is as follows:

- Sections 1- 4 to be submitted within 2 weeks of receipt of the notification
- Section 5 will be completed at the central level (MOHW)
- Section 6 should be completed within 6 weeks of the initial notification
- Section 7 should be completed when the case has completed treatment
- Section 8 to be completed by the MOHW within 6 weeks of completion of treatment

-

A review of the suspected case's medical records will be required with special attention paid to the following:

- Previous History of TB
- Chest X-Ray findings
- Mantoux Test result
- Sputum sample collection
- Sputum results
- HIV testing
- HIV status
- Comorbid conditions (E.g. diabetes, renal disease, etc.)
- Treatment History

Contact Tracing

Contact tracing (WHO, 2012) is a critical component of the case investigation and aims to limit the spread of disease. The investigating officer must:

- Identify all household and bedroom contacts (initiated once notification received)
- Screen all close contacts as follows:
 - Take a thorough history from each contact focused on HIV status, other comorbidities, and recent history of cough, weight loss, and night sweats
 - Conduct a Mantoux test on each household, bedroom, and other close contact
 - Refer as appropriate for further management
 -

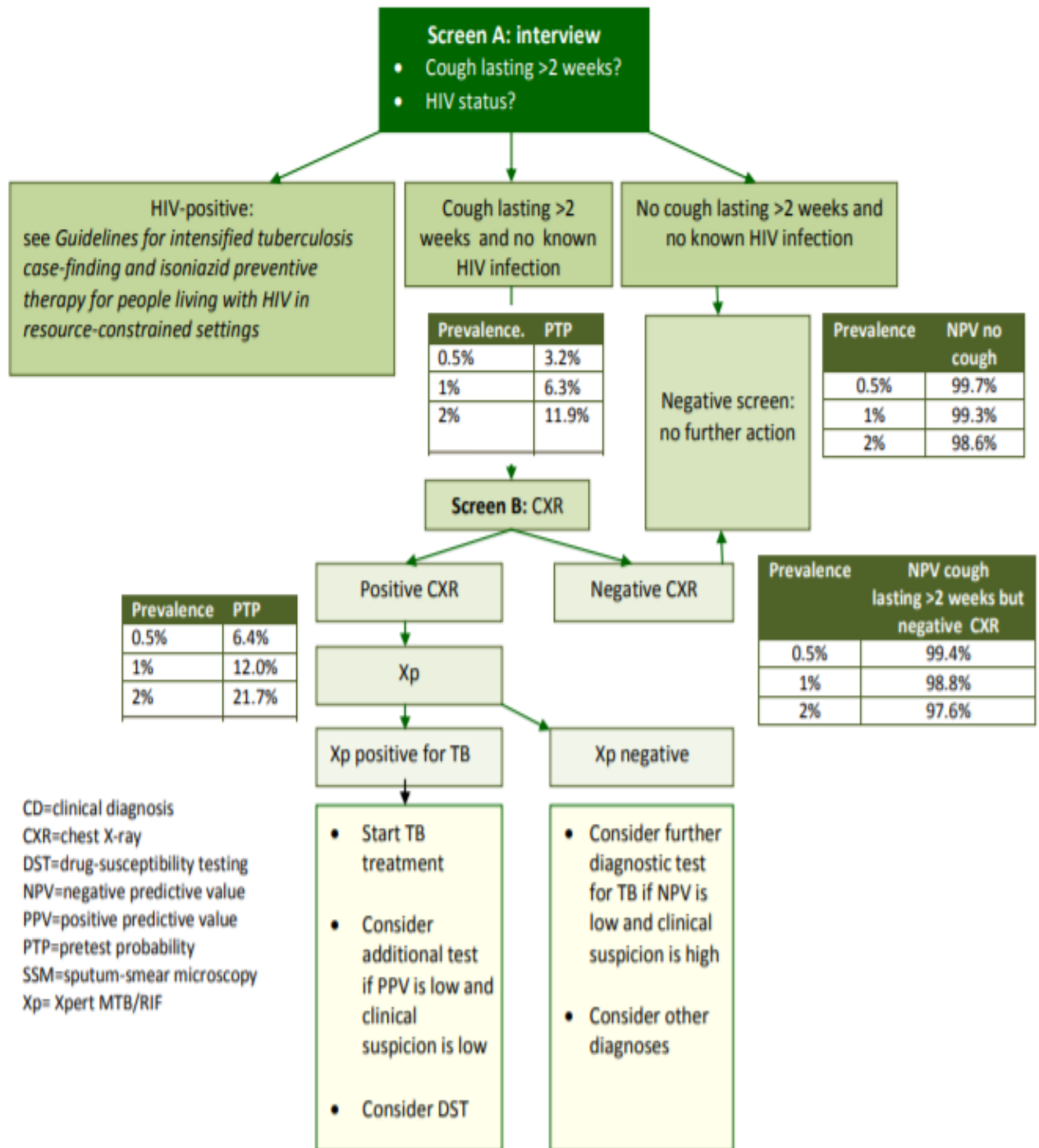
Contacts are prioritized for investigation based on:

1. **Closeness of contact.** For example, persons sharing a bedroom and/or a bed with a clinically diagnosed or bacteriologically confirmed individual with Tuberculosis are more likely to become infected. This is not limited to the household but may include homeless shelters and other settings with multi-person 'wards,' such as an infirmary or areas of confinement such as prisons or lock ups.
2. **Duration of contact.** The longer a person spends in close contact with a person suspected or proven to have Tuberculosis (as above), the greater the risk of transmission and infection.
3. **Immune status of the contact.** A person living with HIV is more likely to progress rapidly to active TB disease once infected.
4. **Presence of pulmonary Tuberculosis in the index case** (the case around whom the investigation is centered). Transmission of disease is more likely in a person suspected or proven to have Pulmonary Tuberculosis than from a person with Extra-Pulmonary Tuberculosis.
5. **Age of the Contact.** Children < 5 years old are more likely to develop severe disease.

Management of Contacts

The management of contacts will depend on the results of initial screening and on the immune status of the contact (Figure 7.4.1).

Figure 7.4.1: Algorithm for Screening of Contacts for Active Tuberculosis
(WHO 2013)



7.5 Data Management

Collection

- A Class 1 form is to be submitted by the medical practitioner to the Parish Health Department or to the National Surveillance Unit within 24 hours of suspicion of a diagnosis of Tuberculosis.
- Sections 1 – 4 of the Tuberculosis Case Investigation Form (Appendix I) are to be completed and submitted within 2 weeks of notification.
- Section 5 will be completed at the central level (MOHW).
- Section 6 of the Tuberculosis Case Investigation Form should be completed within 6 weeks of the initial notification.
- Section 7 should be completed when the case has completed treatment.
- Section 8 to be completed by the MOHW within 6 weeks of completion of treatment.

Collation

- Parishes, Regional Health Authority and the National Surveillance Unit will maintain databases of all notified, suspected, and confirmed Tuberculosis cases.

Reporting

- Parishes will submit to the regional and national levels, case investigation updates as additional data become available.

Analysis

- An annual Tuberculosis report is to be completed by the National Surveillance Unit, in keeping with the epidemiological requirements of the country and the reporting framework of the WHO.

Data Quality

- Biannual TB reports to be obtained from each Regional Health Authority containing the number of notified, confirmed, and co-infected TB cases by age-group and sex.
- Biannual data validation checks are to be carried out by the National Surveillance Unit in keeping with part B of the *Standards and benchmarks for tuberculosis surveillance and vital registration systems* (WHO).

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CHAPTER: VIII

LABORATORY PROCEDURE GUIDELINES

CHAPTER: VIII

LABORATORY PROCEDURE GUIDELINES

- 8.1 Tuberculosis Laboratory Network
- 8.2 Specimen Collection
- 8.3 Specimen Handling Storage and Transport
- 8.4 Smear Microscopy
- 8.5 PCR, Culture and, Drug Sensitivity Testing (DST)

8.1 Tuberculosis Laboratory Network

There is a TB laboratory network that consists of three levels:

- a) Peripheral/Hospital Laboratories
- b) National Public Health Laboratory
- c) Reference Laboratory

The responsibilities of the peripheral hospital laboratory are:

1. Maintain a Biosafety Level 2
2. Ziehl Neelson (ZN) staining of smears and recording results.
3. Participate in internal and external quality control
4. Notification to Parish Health Department of all positive results
5. Maintain laboratory TB register
6. Prepare reports (monthly, quarterly and annual) and submit to NPHL
7. Refer smears to NPHL for validation

The peripheral hospital laboratories are:

- Mandeville Regional Hospital
- St. Ann's Bay Hospital
- Cornwall Regional Hospital
- University Hospital of the West Indies

The responsibility of the National Public Health Laboratory

- Maintain a Biosafety Level 3
- Prepare concentrated or direct smears for ZN microscopy and record results
- The identification of Mycobacteria tuberculosis complex (MTBC) and drug susceptibility testing
- Identification of mycobacteria other than *M. tuberculosis* (MOTT)
- Support training of peripheral laboratory staff in microscopy
- Participate in internal and external quality control
- Provide external quality control for peripheral laboratories
- Provide drug susceptibility testing to monitor treatment programme
- Document cure at end of treatment

The responsibility of the Reference Laboratory

- Maintain a Biosafety Level 3
- Support training of laboratory staff at all levels
- The identification of Mycobacteria tuberculosis complex (MTBC) and drug susceptibility testing through PCR or liquid or solid culture methods

- Identification of mycobacteria other than *M. tuberculosis* (MOTT) and drug susceptibility testing through liquid or solid culture methods
- Provide drug susceptibility testing to monitor treatment programme
- Provide external quality control
- Collaborate with a WHO recognized Supranational Laboratory for external quality assessment (EQA)
- Support research and molecular typing

Samples

- a. Types of samples collected include:
 - i. Sputum or other respiratory samples
 - ii. CSF, spinal / para-spinal / intra-cerebral material*
 - iii. Gastric washings
 - iv. Lymph node or other tissue samples or tissue fluids*
 - v. Blood or bone marrow*
 - vi. Bone*
 - vii. Urine
 - viii. Faeces (fresh or frozen)

* Refer to the National Public Health Laboratory for guidance on collection of sample and suitability.

8.2 Specimen Collection

Proper specimen collection, storage and transport are critical in order to produce laboratory results that clinicians can rely on for definitive diagnosis and treatment decisions.

Sputum or other appropriate specimens should be collected from the affected site as detailed in Table 8.2.1 below. Contact the NPHL for guidance in appropriate specimen collection and transportation.

Sputum Sample Collection Process

The first specimen is taken before the patient leaves as he/she may not return and this may be the only opportunity to do a smear. The second specimen is collected by the patient on arising the next morning before eating. The patient should be instructed to rinse the mouth with plain water, no antiseptic mouthwashes or toothpaste. Sputum is coughed up from deep in the chest and deposited in the container. **Spit, saliva or post-nasal discharges are not acceptable.**

Important details to note for specimen collection include:

- Ensure that no one is standing in front of the patient producing the sputum.
- Explain to the patient that spit or saliva is not suitable and to avoid contaminating the outside of the container with sputum (if the outside of the container is contaminated, discard and repeat procedure using a fresh container).
- Patients should be requested to rinse out their mouths first with water, if they have been chewing food immediately before sputum collection.
- Ask patient to cough deeply, clear the back of the throat and produce about 5-10ml of sputum in the container. Repeat the procedure until a sufficient amount of sputum is obtained.
- Patients should not use antiseptic mouthwashes immediately prior to providing a sputum specimen.

Techniques used when sputum is difficult to obtain include:

1. Bronchoscopy specimens
2. Gastric washings - particularly important for children
3. Sputum induction - particularly important for children
4. Laryngeal swab (not ideal)

These techniques, however, should only be conducted in hospital settings.

Table 8.2.1: Specimen Collection and Transport for Laboratory Testing

Specimen Type	Specimen Requirements	Special Instructions	Unacceptable Specimens
Sputum	Collect 3 specimens (spot, morning, spot) consisting of 5 – 10ml each in a sterile screw cap, leak proof container. Refrigerate or transport rapidly to prevent overgrowth of normal flora.	Patient should rinse mouth with water before collecting sputum to minimize the contamination. Aerosol (saline) induced specimen should be performed only in enclosed area with precautions taken to avoid exposure. The patient inhales slowly and deeply through the mouth and then coughs at will, vigorously and deeply. Sputum is collected in a sterile container.	Pooled specimens. Food particles, residues and other extraneous matter. Specimen in waxed containers may produce false positive results. Unrefrigerated specimens more than 24 hours old may produce contaminated results when cultured.
Tracheal or laryngeal aspirates, bronchial lavage	Collect 3 specimens from patient. Submit 5 – 10 ml in sterile screw cap container. Refrigerate or transport rapidly to prevent overgrowth.	These specimens are collected by trained personnel.	Pooled specimens. Food particles, residues and other extraneous matter. Specimen in waxed containers may produce false positive results. Unrefrigerated specimens more than 24 hours old may produce contaminated results when cultured.

Specimen Type	Specimen Requirements	Special Instructions	Unacceptable Specimens
Gastric Lavage	Submit 10mls or more in a sterile screw cap container; without any preservative; neutralized if more than 4 hours before specimen will be processed.	Collect 3 consecutive days soon after patient awakens and while fasting in order to obtain sputum during sleep. Use sterile saline for lavage. Neutralize the specimen 100 mg sodium carbonate, if time between collection and processing will exceed 4 hours.	Specimen over 4 hours old that is not neutralized.
Body fluids - spinal, pleural, pericardial, synovial, ascitic, blood, pus, marrow and aspirates	Submit at least 10-15ml in a sterile screw cap container, without any preservative.	Disinfect site with alcohol if collection is made with syringe and needle. Blood collected with heparin or sodium polyanethol sulfonate may be inoculated into appropriate blood culture medium for detection of mycobacteria.	Coagulated blood or blood collected in EDTA which inhibits mycobacterial growth.
Tissue	Submit 1g tissue without fixative or preservative . Place in a sterile container and transport immediately to the NPHL.	Collect the specimen aseptically. Do not immerse the specimen in saline or wrap in gauze. Do not freeze.	Specimens in formalin.

Specimen Type	Specimen Requirements	Special Instructions	Unacceptable Specimens
Urine	Submit a clean catch urine in sterile screw cap container. Collect either a single early morning voided midstream sample or the total first morning specimen. Multiple single specimens may be required to optimize yield.	Wash external genitals before voiding to minimize contamination. Specimen should be immediately refrigerated.	Specimen contaminated with faecal matter. Catheter urine.
Wound Skin lesions (tissue) and aspirate	No specific volume requirements.	Skin should be cleaned with alcohol before aspiration of material with syringe. Pus or exudate may be used if volume of tissue is inadequate.	Specimen in Amies, Stuarts transport medium.

(PAHO 2010)

Labelling of Specimens and Forms

Ensure containers and requisition forms are labelled and completed as detailed below.

Label each container with:

1. Patient's name
2. Patient's age
3. Patient's registration number
4. Specimen type
5. Date of collection
6. Hospital or clinic name

Information to be provided on requisition forms:

1. Patient's name

2. Patient's age
3. Patient's registration
4. Sample number
5. Specimen type
6. Date of collection
7. Institution address
8. Parish
9. Referring physician
10. Clinical data
11. Treatment status
12. HIV status

8.3 Specimen Handling and Transportation

Specimen containers must be securely tightened, properly labeled and placed in biohazard specimen bags.

The following procedures must be applied:

- Wear protective gear during receipt and inspection of incoming specimens
- Ensure that sample corresponds with information provided on form
- Inspect the container for signs of leakage. If mass leakage is evident discard the container by autoclaving.
- Disinfect the outside of the container with a suitable disinfectant (10% bleach)
- Record specimen details in appropriate laboratory records

All invasive specimens such as bone marrow and other tissue, must be transported to the NPHL immediately. Sputum specimens are taken to the laboratory on the same day to prevent contamination, if there is a delay it should be refrigerated at 2-8⁰ C and sent to the laboratory within 24-48 hours. Each specimen should be accompanied with a Tuberculosis Laboratory requisition form which is fully completed. Ensure the containers are securely tightened and placed in biohazard specimen bags. Specimens should not be wrapped in forms. The outside of the specimen container may be contaminated with sputum and should be handled with caution. The outside of the contaminated container may be wiped with bleach solution.

All specimens must be transported upright at 2-8⁰C.

8.4 Smear Microscopy

The Ziehl-Neelsen is a simple stain that detects acid fast bacilli (AFB) and can be used to identify positive smears in all laboratories. Sputum smear microscopy for tuberculous bacilli is positive when there are at least 10,000 organisms per ml of sputum. The sputum smear positivity rates in TB/HIV co-infected patients depend on the degree of immunocompromised.

Slide reporting

The number of bacilli seen in a smear reflects disease severity and patient infectivity. Therefore, it is important to record the number of bacilli seen on each smear. Table 8.4.1 below shows the standard method of reporting using 1,000 x magnification for Ziehl-Neelsen smears.

Table 8.4.1: Standard Method of Reporting Using 1,000 x Magnification for Ziehl-Neelsen Smears

Number of Bacilli	Microscopy Fields	Result Reported
Nil	AFB per 100 oil immersion fields	0
1 – 9	AFB per 100 oil immersion fields	Actual or number of AFB seen
10 - 99	AFB per 100 oil immersion fields	+ (1+)
1-10	AFB per oil immersion field	++ (2+)
> 10	AFB per oil immersion field	+++ (3+)

Laboratory technicians should examine all three sputum samples from presumed TB case. Results as indicated in Table 8.4.1 are made available to the clinician and the MOHW Surveillance Unit within 24 hours.

8.5 PCR, Culture, and Drug Sensitivity Testing (DST)

Mycobacterial PCR and culture are more sensitive than smear microscopy and provide a definitive diagnosis of TB. They are therefore seen as the gold standard for bacteriological confirmation.

The modality for identification and DST of MTBC is dependent on burden of disease. The GeneXpert test is a PCR test for TB, which detects the presence of TB DNA as well as testing for the resistance to Rifampin. The GeneXpert is used to diagnose individuals suspected of having MDR-TB, HIV associated TB, all paediatric and new cases (See Appendix IV).

Solid or liquid culture can be utilized for second line drug susceptibility testing and provide bacteriological confirmation of drug resistant TB. Specimens for DST will be referred to a reference laboratory or supranational laboratory.

The final report should be disseminated within one working day of identification being complete.

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CHAPTER: IX

MONITORING AND ADHERENCE TO TREATMENT

CHAPTER: IX

MONITORING AND ADHERENCE TO TREATMENT

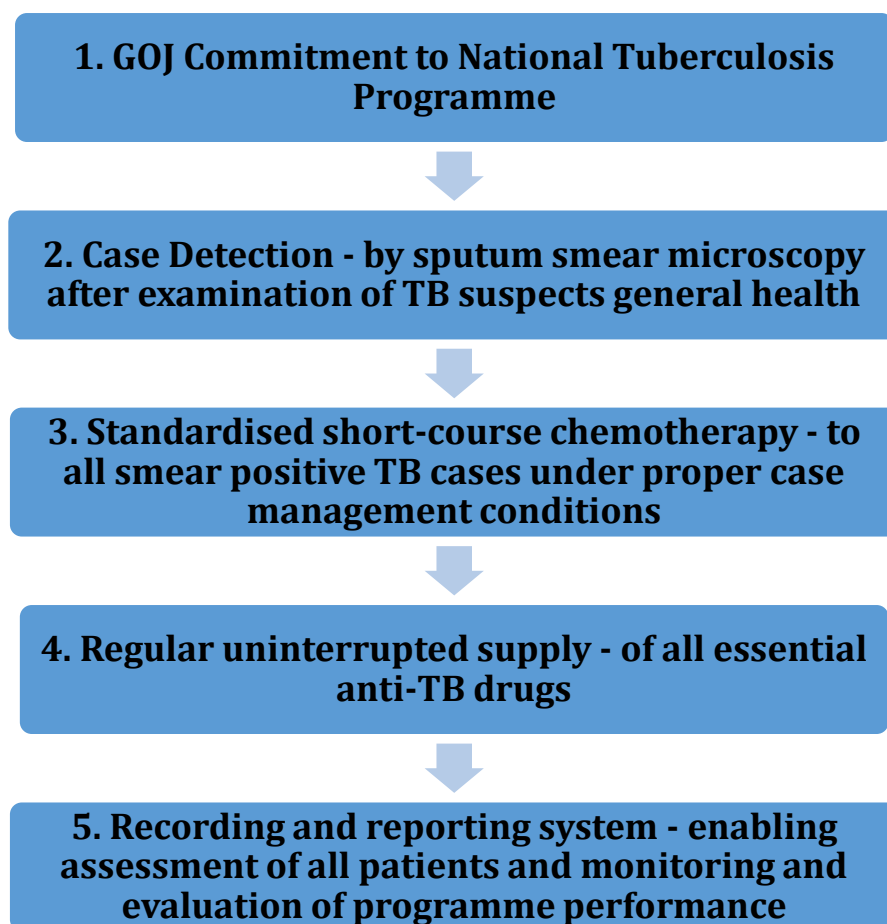
- 9.1 DOTS Five Point Strategy
- 9.2 Procedure for Treatment Monitoring- In Hospital
- 9.3 Procedure for Treatment Monitoring- During Continuation
Phase
- 9.4 Expected Treatment Outcome in TB Cases
- 9.5 Recording and Reporting
- 9.6 Cohort Analysis Treatment Outcome of TB Patients
- 9.7 Key Targets

9.1 DOTS Five Point Strategy

The acronym "DOT" means Directly Observed Treatment, in which a patient is monitored to ensure that the tablets are taken. This ensures that a TB patient takes the right drugs and the right dosage at the right intervals. Jamaica uses a modified approach to DOTS in TB management. A summary of this is outlined in Figure 9.1.1 below

DOTS is conducted in the initial phase of treatment in the hospital and in the continuation phase that is completed in the community.

Figure 9.1.1: DOTS Five Point Strategy



Directly observed treatment is necessary in all phases of TB treatment:

- **Initial Phase** - During the two-month intensive phase of all new smear positive cases, four drugs (Rifampin, Isoniazid, Pyrazinamide, Ethambutol - RIPE) are administered.
- **Continuation Phase** - During the four-months continuation phase for all new smear positive cases two drugs (Rifampin and Isoniazid) are used.
- **Re-treatment** - Throughout the entire re-treatment regimen, RIPE are administered for two months, 2 drugs (Rifampin and Isoniazid) for four months.

DOTS should be adapted to patient needs and working conditions of health care workers to avoid treatment interruptions.

9.2 Procedure for Treatment Monitoring - In Hospital

1. Patients should be educated about DOTS.
2. Confidentiality must be ensured so that direct observation of treatment is acceptable to the patient.
3. After producing a negative smear, patients should be discharged from the hospital with a prescription for two month's supply of anti-TB medication. **The medication should be free of charge.** At the time of discharge, patients get a two month's appointment date to return for medical assessment.
4. Prior to discharge, a meeting should be held between patient, relatives, preferably primary and secondary health care workers, to discuss the continuation phase. Issues to be discussed include:
 - Need for a supervisor
 - Responsibility of the patient
 - Frequency of follow-up
 - Potential problems the patients may face during treatment
 - Designated health care liaison

9.3 Procedure for Treatment Monitoring - During the Continuation Phase

1. The hospital should prepare a discharge summary in triplicate and copies sent to the following:
 - Patient docket
 - Patient
 - Relevant parish health department

A treatment card is provided to the patient. Home visits by the public health nurse and hospital visits by the patient are documented on the card.

2. The discharged patient should report to the nearest health centre to his home with the discharge summary.
3. The person in charge of the health centre should register the patient for management and send a Class 1 notification to the health department.
4. A DOTS observer is to be assigned to the case (if not yet done) to monitor the continuation phase. A meeting is to be held with the patient and relatives to decide on the assignment of the DOTS observer.
5. Each patient should be observed taking medication daily or a minimum of three times weekly.
6. The DOTS Observer must complete a daily DOT log. The log consists of the patient's name and address, medication and schedule (observed, not observed, not available), time, side effects, other comments and the health care worker's initials.
7. The DOTS Observer must count the remaining tablets to confirm that the reported number of tablets taken is correct.
8. Patients should be reminded about medical outpatient appointments and assisted to make transportation arrangements (if indicated).
9. Patients and family should be educated about taking the medication as prescribed.
10. The DOTS Observer should explain and arrange for the patient to provide sputum samples at designated intervals to monitor treatment responses. Arrangements can include provision of sputum specimen containers, explanation of collection process, storage and transportation to the laboratory. This is outlined in Table 9.3.1 below,

Table 9.3.1: Sputum Smear Requirements for Treatment Regimen

Sputum Smear Examination	Treatment Regimen	
	Six month regimen	Eight month regimen
At end of initial phase	End of second month	End of second month
In continuation phase	End of fourth month	End of fifth month
At end of treatment	In the sixth month	In the eighth month

Source: CAREC TB manual of prevention and control procedures (1997)

11. Enquiries should be made about any incidence of adverse drug reaction. Any adverse reactions should be reported through the MOHW pharmacovigilance form available at the MOHW website (See Appendix V) and submitted to the MOHW.
12. Patients reporting adverse drug reactions should be referred to the doctor.
13. Treatment defaulters should be located and encouraged to return.
14. The DOTS observer must report to the public health nurse in charge of the case concerning the patient's adherence to treatment, on a minimum monthly basis.
15. The DOTS observers must complete and submit the TB treatment card to the public health nurse.
16. The DOTS observer must report any abnormal findings to the public health Nurse within 24 hours.

9.4 Expected Treatment Outcome in TB Cases

The treatment outcomes for TB are detailed in Table 9.4.1

Table 9.4.1: Expected Treatment Outcomes in TB cases

Treatment Outcome	Description
Cured	A pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who was smear- or culture-negative in the last month of treatment and on at least one previous occasion.
Treatment Completed	A TB patient who completed treatment without evidence of failure BUT with no record to show that sputum smear or culture results in the last month of treatment and on at least one previous occasion were negative, either because tests were not done or because results are unavailable.
Treatment Failed	A TB patient whose sputum smear or culture is positive at month 5 or later during treatment
Died	A TB patient who dies for any reason before starting or during the course of treatment.
Loss to Follow-up	A TB patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more.
Not Evaluated	A TB patient for whom no treatment outcome is assigned. This includes cases “transferred out” to another treatment unit as well as cases for whom the treatment outcome is unknown to the reporting unit.
Treatment Success	The sum of patients who are cured and those who have completed treatment

9.5 Recording and Reporting

Accurate record keeping on all patients, maintenance of registers and reporting as per schedule are minimum requirements.

Reports are used for monitoring, evaluation, planning and budgeting purposes. Reports are generated from the tuberculosis registers and field records. At each level of the treatment monitoring process, a treatment register should be established to collect data for cohort analysis. It is important that all registers are kept updated with complete, accurate and reliable data.

List of Monitoring Records

The areas listed must maintain a Tuberculosis Register:

- Laboratory Register
- Health Centre
- Hospital
- District/Zone
- Parish
- Regional
- National

The following forms and reports are required:

- TB Sputum Culture Request Form
- Class 1 Notification Form
- TB Case Investigation Form
- Contact tracing log book
- Quarterly case notification reports
- Quarterly report on treatment - cohort analysis

Each parish should maintain a TB register (Appendix VI). The register should have a section for treatment outcomes. On submission of the reports from the health centre and district/zones, the reports should be compiled by the Parish TB Coordinator on a monthly, quarterly and annual basis and submitted to the Medical Officer of Health.

The district quarterly case-finding report and treatment outcomes are submitted through the parish MOs(H) to the regional epidemiologist/surveillance officer, National Surveillance Unit and the National TB Programme.

Presently the TB Parish registers are used to monitor patient's adherence to treatment schedule.

With the aim of developing a feasible local system, there are plans to carry out the following:

- Monitoring of patient usage and the development of a formalized DOTS Programme to take appropriate action.
- Exploring linkages with the use of adherence counsellors (HIV/ AIDS Programme).

9.6 Cohort Analysis/Treatment Outcome of TB Patients

Cohort analysis on treatment outcomes should be done every quarter and at the end of the year. A Cohort of TB patients consists of all the sputum smear positive or smear negative **and** Xpert positive (as in HIV/TB coinfection) pulmonary cases registered during a quarter.

New and previously treated patients form separate cohorts. An evaluation at the end of treatment (6 or 8 months) outcome takes place about three (3) months after all patients in the cohort have completed their course of treatment.

9.7 Key Targets

Targets of the WHO END TB strategy to end the global TB Epidemic include:

1. Reduction in the number of TB deaths by 95% by 2035 compared with 2015.
2. Reduction in TB incidence rate by 90% by 2035 compared with 2015.
3. By 2035 to have 0% of TB affected families facing catastrophic costs due to TB.

WHO Global Priority Indicators to Monitor END TB Strategy

Ending the TB epidemic is included in the United Nations Sustainable Development Goals (SDGs) by 2030 under Goal 3. To monitor the implementation of the END TB strategy, WHO recommended the use of the global priority indicators and targets (WHO 2015) outlined in Table 9.7.1, of which all countries should aim to reach by 2025.

Table 9.7.1: WHO Global Priority Indicators to Monitor the END TB Strategy

Global Priority Indicator	Definition	Target
1. Treatment Coverage	Number of people that developed TB, and were notified and treated, out of the total estimated number of incident cases in the same year (%).	≥ 90%
2. TB Treatment success rate	Number of TB patients who were successfully treated out of all notified TB cases (%).	≥ 90%
3. Preventive treatment coverage	Number of people living with HIV and children who are contacts of cases who were started on preventive treatment for latent TB infection, out of all those eligible (%).	≥ 90%
4. TB affected households facing catastrophic costs*	Number of TB patients and their households that experienced catastrophic costs due to TB, out of all TB patients (%).	0%
5. Uptake of new diagnostics and new drugs*	Number of TB patients who were diagnosed using WHO-recommended rapid tests, out of all TB patients (%). Number of TB patients who were treated with regimens including new TB drugs, out of those eligible for treatment with such drugs (%).	≥90%

* Not currently reported on. Measure being reviewed nationally.

Guideline: Monitoring and Adherence to Treatment		
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CHAPTER: X

PROCUREMENT – DRUGS AND LABORATORY SUPPLIES

CHAPTER: X

PROCUREMENT – DRUGS AND LABORATORY SUPPLIES

10.1 The Drug Logistic Cycle

10.2 Choice

10.3 Purchasing

10.4 Management of Stocks

10.5 Distribution of Storage

10.2 Choice

Choice is dependent on:

- Information coming out of laboratories - TB strains and what are the drug resistant strains if any.
- Efficacy of available drugs

Estimated Quantity of Drugs and Laboratory Supplies

The quantity of drugs, reagents and supplies is estimated based on:

- The target population which is estimated using the TB register and the Hospital Active Surveillance (HAS)
- National treatment protocol algorithm
- Trend analyses

Monitoring

Monitoring (responsibility of the clinicians and the laboratory) with feedback to the relevant stakeholders (MOHW Standards and Regulations Division) of drug effectiveness is necessary to ensure appropriate preventative and corrective action (See Appendix VIII – Pharmacovigilance Form).

10.3 Purchasing

The National Health Fund (NHF) is responsible for the purchasing of drugs.

Laboratory supplies are procured through the NPHL local laboratory budget. Purchasing of drugs and laboratory supplies will be according to GOJ guidelines and timelines with respect to the tender process using the restricted tender method. The quality of drugs must be maintained and must meet the standards set by the Standards and Regulations Division of the MOHW. Budgetary allocations must be ensured under GOJ financial arrangements (budgetary and extra budgetary).

Close collaboration between NTP manager and procurement officers is important. It must be ensured that the national drug formulary contains the essential drugs (both first and second line drugs) and is regularly updated. Currently not all second line drugs are included in the national drug formulary.

A separate tender for anti-tuberculosis drugs is performed, because:

- The demand comes from a programme that has precise conditions.
- To ensure that drugs are supplied in the required quantity and time frame, thus allowing patients to follow their treatment as recommended.
- There are very few manufacturers of quality drugs, and a restricted tender is suitable.
- The drugs are given free of charge and receive specific funding, and a separate tender facilitates their management.
- The tender is the final point of a process that involves several different public departments, all of whom have a say in the final result.

Monitoring

The tender process must be closely monitored and post marketing surveillance must be carried out.

10.4 Management of Stocks

Consistent monitoring of stock is essential. A reorder level based on usage rate, lead time, minimum and maximum stock levels, buffer/ safety stock must be implemented.

Constant liaison between purchaser and NTP as well as with relevant ministries is necessary to maintain adequate stock levels. The management of anti-tuberculosis drugs is based on theoretical consumption, i.e., it is linked to the number of tuberculosis patients on treatment during a defined time period.

The quantity of anti-tuberculosis drugs that needs to be ordered depends on several factors:

- The number of tuberculosis patients detected and notified in the *HAS*
- The projected number of patients to be diagnosed for the period being quantified
- The standardised treatment regimens used and set out in the NTP manual, and the fixed-dose combinations used
- The quantity of drugs currently stocked at the central level
- The reserve stock required
- The drug consumption from the moment the order is made to the moment it arrives (this depends on the transport time)
- The frequency of the ordering
- The budget

10.5 Distribution and Storage

The following factors are very important:

- Appropriate storage conditions should be maintained - temperature and humidity
- Control of expiry date
- Reserve stock
- Implementation of first expired first out (FEFO) principle
- Control of handling and transport at all levels
- Implementation of drug accounting system at all levels (where drugs are stored or administered)

Where feasible, pre-packaging in patient specific dosages is advised to avoid wastage.

Guideline: Procurement – Drugs and Laboratory Supplies		
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CHAPTER: XI

ROLES AND RESPONSIBILITIES OF NTP PERSONNEL

CHAPTER: XI

ROLES AND RESPONSIBILITIES OF NTP PERSONNEL

11.1 Parish/Local Level

11.2 Regional Level

11.3 National Level

11.1 Parish/Local Level

The implementation of NTP policies and guidelines occur at the service level.

All Health Care Workers

- Suspect and detect Tuberculosis among patients reporting for health care and take all necessary and appropriate measures for the examination of those patients.
- Educate and examine household contacts of patients diagnosed with Tuberculosis.
- Manage patients upon instructions from the clinician/medical officer and the Tuberculosis coordinator.
- Encourage and supervise and/or arrange the supervision of patients for directly observed therapy.
- Educate Tuberculosis patients on the need for regular attendance and on the importance of adherence to treatment.
- Ensure that treatment progress is monitored periodically, according to the NTP guidelines.
- Identify patients who need to be referred to a medical officer with regards to complications of Tuberculosis, according to the guidelines of the NTP.
- Maintain the Tuberculosis register at the various levels of the health service (health centre, hospital, laboratory, district, parish and regional).
- Locate and return to treatment those patients who are irregular in attendance.
- Ensure that records of the attendance, treatment and sputum results are kept in the appropriate registers and files.
- Conduct health education to new patients and their household contacts on the importance of early diagnosis and the possibilities of cure.
- Participate in public education on Tuberculosis and its management.
- Any other duties pertaining to the implementation of the national TB programme guidelines

Hospital Staff

- Notify suspected cases as Class 1 Notifiable Disease, on suspicion, to the relevant Health Department and National Surveillance Unit.
- Educate patients, caregivers and family about TB, DOTS, treatment regimens and the possibility of cure with adequate treatment.
- Arrange meeting with patient, relatives, primary and secondary care health workers concerning the continuation phase of treatment in the community.

- Ensure completion of the discharge summary in triplicate (one copy to be sent to the relevant Health Department, one copy to be given to the patient and one copy to be retained in the patient docket).
- Ensure that patient gets medications free-of-charge and is given an appointment date for clinical review.
- Any other duties pertaining to the implementation of the National TB Programme guidelines.

Health Centre/Health District Supervisory Public Health Nurse

- Receive notification of cases in respective areas and enter information in the Parish TB Register.
- Assign an appropriate DOTS Observer to each case taking into account decisions taken at the meeting with relatives and clinicians.
- Supervise DOTS Observer's functions through review of Treatment Cards, field visits, etc.
- Provide monthly reports to the Parish TB Coordinator (on notifications, investigations, monitoring of cases, DOTS, etc.).
- Ensure case investigation and contact investigation are performed, and submit relevant reports within time lines set out in NTP Guidelines.
- Ensure adequate supplies of personal protective equipment (PPE) for suspected case/contact investigation.
- Any other duties pertaining to the implementation of the national TB programme guidelines.

DOTS Observer

- Educate patients about TB, DOTS and the importance of compliance to treatment guidelines
- Monitor the taking of medication by cases through direct observation.
- Ensure accurate completion of the individual TB Treatment Card.
- Supervise adherence to medication through tablet counts and other methods.
- Make note of and report any incidences of adverse drug effects to the PHN.
- Report weekly to PHN on the adherence of the cases to the continuation phase regimen.
- Ensure that patient keeps medical appointments and fulfils laboratory and other requests.
- Submit completed treatment card to health centre or public health nurse at completion of continuation phase of treatment.
- Locate and persuade defaulters to return to treatment.

- Any other duties pertaining to the implementation of the national TB programme guidelines.

Tuberculosis Parish Coordinator

The Parish TB Coordinator may have to divide his/her time between TB programme activities and those for other programmes under their responsibility.

- Ensure entry of case in Parish TB Register and maintain and update register monthly.
- Assign public health nurse and public health inspector to do community case investigation, noting date assignment made and expected date for report submission, in keeping with national guidelines.
- Ensure that the assigned public health personnel appoint a DOTS Observer to each case.
- Ensure the establishment of the TB programme structure in health centres, health districts and hospitals of the parish.
- Collate and review case investigation and other reports on a weekly basis for submission to the Medical Officer (Health) within the timeline indicated in the Guidelines (TB Case Investigation Form).
- Set targets, in collaboration with the Medical Officer (Health), for TB programme performance in the parish, and monitor indicators.
- Compile monthly, quarterly and annual reports for submission through the Medical Officer (Health) to the Regional Technical Director/Regional Epidemiologist/Regional Surveillance Officer.
- Arrange training of parish staff, with regards to requirements of NTP Guidelines.
- Collaborate in the arrangements for procurement of drugs and supplies for the TB programme.
- Ensure that TB issues are discussed at parish technical and epidemiology committee meetings.
- Any other duties pertaining to the implementation of the national TB programme guidelines.

Medical Officer (Health)

At the parish level, the Medical Officer (Health) is responsible for planning, implementation, monitoring and evaluation of the TB programme.

- Ensure the development of an annual work plan for TB programme management (including procurement, financing/budgeting, and establishment of targets in keeping with national performance targets).

- Ensure that TB programme is implemented at parish level in keeping with NTP policies and guidelines.
- Monitor TB programme planning, implementation and evaluation activities of the parish TB coordinator.
- Check and endorse all Class 1 Disease Notifications including TB notifications.
- Conduct field visits at hospitals to ensure Hospital Active Surveillance is conducted.
- Review all case investigation reports for quality, completeness and timeliness.
- Collect and analyze all monthly, quarterly and annual reports submitted by parish TB coordinator, prior to submission to Regional or National Level.
- Monitor the TB programme for achievement of targets and indicators.
- Collaboratively formulate strategies to address issues impacting the implementation of the TB programme in the parish.
- Reconcile parish register along with TB coordinator quarterly.
- In collaboration with parish coordinator, conduct quarterly review of the TB programme.
- Submit quarterly reports to the Regional Technical Director (RTD), Regional Epidemiologist and National TB Coordinator simultaneously.
- Ensure adequate supplies/drugs for TB case management in the parish.
- Any other duties pertaining to the implementation of the national TB programme guidelines.

11.2 Regional Level

Regional Epidemiologist/Regional Surveillance Officer

- Plan the regional TB programme in conjunction with MOs(H), in keeping with national priorities and targets/indicators.
- Monitor the regional TB programme's implementation and achievement of targets.
- Compile regional report on TB programme, in keeping with Service Level Agreement and National targets, for submission to RTD and National TB Coordinator.
- Conduct reviews of parish TB programmes, in conjunction with MOs(H).
- Act as liaison between MOs(H)/RTD and National officers in NTP.
- Ensure that training and supervisory support are provided within the region to personnel who perform activities related to TB control, including laboratory personnel.
- Supervise the record keeping of the TB Registers, and the laboratory registers.
- Review for accuracy and completeness all quarterly TB reports compiled in the region and provide analysis and feedback to the district level.
- Collaboratively ensure procurement of supplies of drugs, laboratory reagents, slides, sputum containers, and forms and oversee distribution of supplies to the peripheral health units.
- Any other duties pertaining to the implementation of the national TB programme guidelines.

Regional Technical Director (RTD)

At the regional level, the Regional Technical Director is responsible for planning, financing, implementation, monitoring and evaluation of the TB control programme.

- Liaise with the NTP in relation to policies, implementation, plans, standards and guidelines.
- Ensure that the regional TB programme is implemented in keeping with national goals, objectives, policies and procedures.
- Ensure that the regional TB programme is evaluated twice per year on process/implementation, outputs/outcomes, and impact.
- Submit quarterly reports to NTP on TB control in the Regional Health Authority, in relation to regional, national and international targets and indicators.
- Supervise the monitoring of the TB programme by the Regional Epidemiologist/Regional Surveillance Officer.
- Any other duties pertaining to the implementation of the national TB programme guidelines

11.3 National Level

National Tuberculosis Laboratory

- Preparation and dissemination of the laboratory manual to ensure that adequate policy and standards regarding microbiological diagnosis of *M. tuberculosis* are maintained throughout the country.
- Establish and provide supervision to a network of microscopy centres.
- Facilitate/coordinate training and supervision of the laboratory technicians who perform sputum smear microscopy.
- Facilitate/coordinate training and supervision of the laboratory technicians in implementing a respiratory protection programme.
- Develop and implement a quality control procedure for smear microscopy.
- Provide technical advice on the supply of reagents, laboratory supplies and equipment to improve diagnostics.
- Collate and provide monthly report of all drug resistance of *M. tuberculosis* to the NTP.
- Collaborate in research programme.
- Any other duties pertaining to the implementation of the national TB programme guidelines.

National Surveillance Unit

- Plan the national TB programme in conjunction with MOs(H), in keeping with national and international priorities and targets/indicators.
- Monitor the national TB programme's implementation and achievement of targets.
- Compile the national report on the TB programme, in keeping with national targets and international targets, for submission locally and internationally.
- Ensure the implementation of programme policies and strategies, with regards to planning, implementation, monitoring and evaluation of the NTP in the Region/District/Parish under the technical guidance of the Medical Officer (Health).
- In conjunction with the National TB Diagnostic Committee, assist regional and parish health care workers in tuberculosis case-finding, diagnosis and classification.
- Collaborate with hospital and parish staff to ensure that case investigation and management are in keeping with national standards and guidelines.
- Collect and analyze data on tuberculosis control activities and compile reports on case notifications, treatment outcomes, and provide feedback to TB programme staff.
- Coordinate the management of all TB-related information.
- Maintain the Central TB Register.

- Review for accuracy and completeness all quarterly TB reports compiled in the region and provide analysis and feedback to the district level.
- Any other duties pertaining to the implementation of the national TB programme guidelines

National Tuberculosis Programme

At the national level, the National Tuberculosis Programme Manager is responsible for all aspects of the TB control programme, including development of policies, guidelines and regulations and monitoring and evaluation.

- Supervise the overall function of the national TB control programme.
- Provide leadership, encouragement/motivation and advice to all NTP personnel.
- Work with individuals in the central Ministry of Health and Wellness, regional authorities, individuals from international organizations, donor agencies to plan the NTP strategy.
- Ensure the establishment of programmes in special settings (e.g. prisons).
- Ensure that training, supplies, and supervisory support are provided to personnel at all levels of the health system in the districts where DOTS is introduced for tuberculosis control.
- Initiate and co-ordinate health education and training through learning institutions and on-the-job activities, for health care workers and the general public.
- Ensure the procurement and reserve of drugs, laboratory and other supplies, and ensure their distribution and appropriate use in the health service.
- Maintain regular contact with health facilities, including NGOs, and hospitals to which patients may be referred.
- Ensure problem-solving in relation to NTP at central level.
- Ensure adequate GOJ budgetary allocation to NTP and obtain commitments for extra-budgetary funding.
- In collaboration with national laboratory director, ensure that the national/regional laboratories have capacity for isolation identification and anti-microbial susceptibility testing of *M. tuberculosis*.
- Ensure periodic updating of the national TB programme guidelines.
- To review reports from the national laboratory director, the national surveillance unit and parish TB coordinators and provide feedback to the field and the Senior Medical Officer – HIV/STI/TB Unit.
- Ensure that all NTP staff receive updates in training periodically.
- Facilitate the involvement of NGOs in tuberculosis prevention and control efforts.
- Any other duties pertaining to the implementation of the national TB programme guidelines

Guideline: Roles and Responsibilities of NTP Personnel		
Date Revised: November 2021	Distribution to hospitals and health centres	Index: XI
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CHAPTER: XII

MONITORING AND EVALUATION OF TB CONTROL PROGRAMME

CHAPTER: XII

MONITORING AND EVALUATION OF TB CONTROL PROGRAMME

12.1 Monitoring and Evaluation – Parish Level

12.2 Monitoring and Evaluation – Regional Level

12.3 Monitoring and Evaluation – National Level

12.4 Reporting

12.5 Contents of Monthly Reports

12.6 Programme Targets

Monitoring and Evaluation of TB Control Programme

Monitoring is the routine tracking of the information about the TB Programme. Evaluation is the assessment of the efficiency and effectiveness of the interventions under the Programme.

Monitoring and Evaluation should be done at all levels:

- Parish
- Regional
- National

12.1 Monitoring and Evaluation at the Parish Level

The Parish Coordinator is responsible for monitoring and evaluation at the parish level. The Parish Coordinator needs to ensure that the following activities are undertaken:

- TB notifications meet programme requirements for case investigation
- Records are maintained
- Field Investigations are completed and reports submitted
- Reports are compiled and submitted to regional health authority and the national surveillance unit on a timely basis
- Availability of drugs and supplies e.g. PPD, anti-TB medication and lab supplies (in stock).
- Budget prepared
- DOTS requirements being met - staffing, traveling etc.
- Audits of hospitals - HAS procedure and field visits
- Case investigations reviewed for quality, completeness and timeliness.
- Submission of monthly, quarterly and annual reports e.g. cohort analysis, monthly case findings etc.
- Monitor the implementation of set work plan for achievement of targets and indicators.
- Strategies formulated to combat barriers to success
- Monthly reconciliation of parish register with NTB coordinator
- Develop a LTBI register for the parish
- Quarterly review of the TB surveillance programme
- Conduct quarterly review of treatment outcomes

12.2 Monitoring and Evaluation at the Regional Level

The Regional Technical Director (RTD) plays an important role in monitoring and evaluation of the TB Programme at the Regional Health Authority. The RTD or their designate should ensure that the following activities are undertaken:

- Monitoring of the frequency of submission of reports from the parishes
- Analysis of parish reports for achievements of indicators and targets set out in work plan
- Audit of parish TB surveillance system
- Preparation of budget

12.3 Monitoring and Evaluation at the National Level

The objectives of monitoring and evaluation include:

- To determine the implementation of policy decisions
- To determine the status of implementation of surveillance strategies
- To determine the achievement of targets and indicators set out in the work plan
- To conduct an audit of the TB Programme on a biennial basis

A multi-stakeholder TB Diagnostic Committee should be established with a secretariat and additional key stakeholders. The committee will meet biannually.

The secretariat will include:

- Director TCS HSTU MOHW
- National TB Programme Coordinator – (Programme Development Officer HSTU)
- MO(H) representative from WRHA
- MO(H) representative from SERHA
- Senior Medical Officer of National Chest Hospital

In addition to the secretariat the TB Diagnostic Committee will include the following:

- Kingston Public Hospital Infectious Disease Specialist
- UWI Pulmonologist
- Laboratory
- Representative from HIV Programme
- Nursing representative
- National surveillance unit representative

This TB Diagnostic committee will review the following:

- Hospital treatment
- Community treatment
- Defaulters
- Laboratory and radiological results
- Classification of cases
- Treatment outcomes

The functions of the TB diagnostic committee are supported by the National TB Coordinator. Activities to be undertaken include:

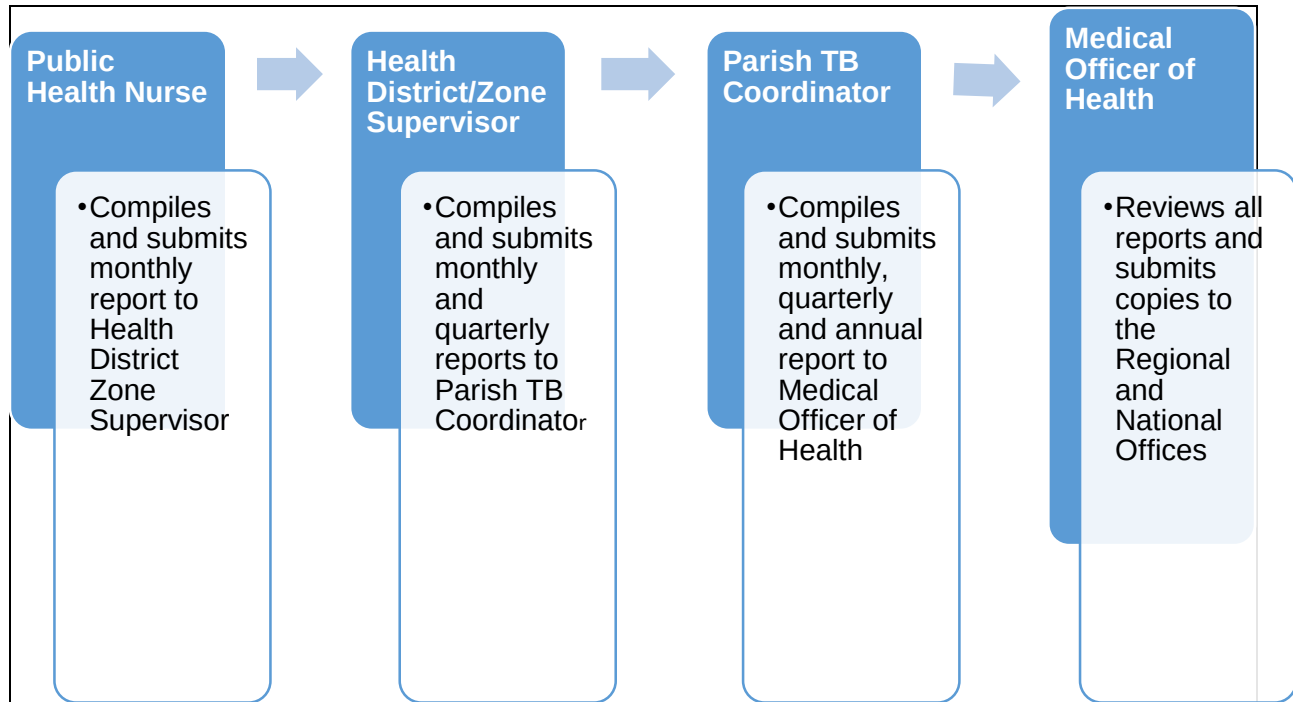
- Preparation of quarterly, annual and special reports as well as process evaluation
- Provide feedback from evaluations
- Checks on data quality

- Analysis of trends in TB
- Utilization of available and appropriate technology for data storage and retrieval
- Research
- Preparation of programme budget

12.4 Reporting

Reporting should follow the lines of communication as shown in Figure 12.4.1 below.

Figure 12.4.1: Reporting Lines of Communication



12.5 Contents of Monthly Reports

- Number of notifications in each health district
- Percentage of notifications confirmed, discarded and under investigation
- Percentage of case investigations initiated, completed and pending
- Number of extra-pulmonary culture positive and negative cases on treatment
- Number of new sputum positive and smear negative cases
- Number of new smear positive cases grouped according to age and sex
- Number of relapses, failures, defaulters and deaths

12.6 Programme Targets

Programme targets for the various indicators are grouped in the following areas:

- Global reporting
- TB/HIV co-infection
- Programme outcomes
- Political commitment and programme monitoring
- Diagnosis and laboratories
- Case management and treatment
- Case and contact investigation
- Drug management
- Recording and reporting
- Supervision
- Human resource development

Guideline: Monitoring and Evaluation of TB Control Programme		
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ANNEXES

Appendix I – Tuberculosis Case Investigation Form

Appendix II - Paediatric Extra – Pulmonary Tuberculosis
Clinical Presentation and Management

Appendix III – Class 1 Notification Form

Appendix IV – GeneXpert Investigation

Appendix V – Pharmacovigilance Form

APPENDIX I

TUBERCULOSIS CASE INVESTIGATION FORM

MINISTRY OF HEALTH and WELLNESS, JAMAICA

TUBERCULOSIS INVESTIGATION FORM

(All dates in the dd/mm/yyyy format)

Facility of notification: _____ Ward: _____

Date of notification: _____

Date Investigation Assigned: _____ Investigating Officer: _____

Medical Record No: _____ Physician/Consultant: _____

N.B: Section 1 to 4 must be submitted to the Parish MO(H) 2 weeks or less after notification date (i.e. date of case of recognition as suspected T.B.)

Section 1 – Demographic Information

Last Name: _____ First Name: _____ Middle Name: _____

Pet name: _____

Sex: Male ☐ Female ☐ Age: ____ (years) ____ (months) DOB (d/m/y): _____

Address

Street: _____

Community: _____

Email : _____

Nationality: _____ Employed Y ☐ N ☐

Name of Employer/School: _____

Employer/School Address: _____

Travel History (past 6 weeks) Y ☐ ☐ ☐ If Yes, Country/Region: _____

Contact

Phone (H): _____

Phone (C): _____

Parish: _____

Occupation: _____

Next of Kin: _____

NOK Phone No: _____

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Section 2 – Clinical and Social Information (Tick as appropriate)

Date of onset of Symptoms:					
Symptom		Duration	Symptom		Duration
Cough	Y ☐ N ☐ U ☐		Chest Pain	Y ☐ N ☐ U ☐	
Fever	Y ☐ N ☐ U ☐		Night Sweats	Y ☐ N ☐ U ☐	
Haemoptysis	Y ☐ N ☐ U ☐		Weight Loss	Y ☐ N ☐ U ☐	
Medical History					
HIV Infection	Y ☐ N ☐ U ☐	Date diagnosed ☐	ARV Therapy	Y ☐ N ☐ U ☐	Date ARV Started:
Diabetes	Y ☐ N ☐ U ☐	COPD	Y ☐ N ☐ U ☐	Renal Disease	Y ☐ N ☐ U ☐
Liver Dysfunction	Y ☐ N ☐ U ☐	Other	Y ☐ N ☐ U ☐	Specify _____	
History of BCG	Y ☐ N ☐ U ☐	BCG Scar Seen	Y ☐ N ☐ U ☐	Co-trimoxazole (Bactrim) Therapy	Y ☐ N ☐ U ☐
Smoking	Y ☐ N ☐ U ☐	Alcohol Use	Y ☐ N ☐ U ☐		
Other Drug Use	Y ☐ N ☐ U ☐		Specify: _____		
Pregnant	N/A ☐ Y ☐ N ☐ U ☐		LMP: YYYY/MM/DD		
Social Assessment (*Social worker to complete after assessment done)					
Living situation and conditions (Tick appropriate response)					
Lives alone ☐ With partner ☐ With friends ☐ Family ☐					
Other ☐ Specify: _____					
Specify number of persons in household: _____					
Type of house: _____ Number of rooms: _____					
Number of persons per room: _____					
Do other persons live on the premises: Y ☐ N ☐ Specify Number: _____					
Social support: Y ☐ N ☐		Name of support: _____			
		Contact number of support: _____			
Financial support for the duration of treatment (source): _____					
Barriers to adherence identified: Y ☐ N ☐ Specify: _____					

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Section 3 – Investigations (Tick as appropriate)

		Date of Study		
Mantoux	Y ☐ N ☐ U ☐		Pos ☐ Neg ☐	Reading (mm) :
X-Ray (type):	Y ☐ N ☐ U ☐		Pos ☐ Neg ☐	Findings:
Sputum 1	Y ☐ N ☐ U ☐		Smear: Pos ☐ Neg ☐	Culture: Pos ☐ Neg ☐
Sputum 2	Y ☐ N ☐ U ☐		Smear: Pos ☐ Neg ☐	Culture: Pos ☐ Neg ☐
Sputum 3	Y ☐ N ☐ U ☐		Smear: Pos ☐ Neg ☐	Culture: Pos ☐ Neg ☐
HIV Test	Y ☐ N ☐ U ☐		Pos ☐ Neg ☐	
Gene-Expert	Y ☐ N ☐ U ☐		Pos ☐ Neg ☐	Resistance Profile: RIF: Y ☐ N ☐

Section 4 – Treatment Initiation – Intensive Phase (Tick as appropriate)

Anti-TB Treatment: Y ☐ N ☐	Date Started:	Date Completed:
Treatment Site (Name of Hospital):	Med. Records No:	
Supervision-DOT: Y ☐ N ☐	Duration in Hospital:	Date of Discharge:
Drugs Used:	Dosage:	Duration:
Isoniazid ☐		
Rifampicin ☐		
Pyrazinamide ☐		
Streptomycin ☐		
Ethambutol ☐		
Other (Specify) ☐		

Date Form Completed: _____

Name of Investigator: _____ Signature: _____

Parish MO(H): _____ Signature: _____

Parish MO(H) Comment: _____

_____ Date: _____

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Section 5 – Case Classification (Tick as appropriate)

To be completed by the Ministry of Health and Wellness

Classification: Bacteriologically Confirmed ☐ Clinically Diagnosed ☐ Discarded ☐		
Classification date:		
Drug Resistance: Nil ☐ Monoresistance ☐ Polydrug Resistance MDR ☐ XDR ☐ RR ☐		
MDR – Multi-drug resistant TB; XDR - Extensively Drug Resistant TB; RR – Rifampicin Resistant TB		
Isoniazid ☐	Rifampicin ☐	Ethambutol ☐ Pyrazinamide ☐
Treatment History:		
1. New Case ☐		
2. Previous TB Treatment ☐	a. Relapse	☐
	b. Treatment After Loss to Follow-up	☐
	c. Treatment After Failure	☐
	d. History Unknown	☐
	e. Other	☐
3. Previous TB Treatment History Unknown ☐		
Anatomical Site:		
Pulmonary Tuberculosis ☐	Extra-Pulmonary Tuberculosis ☐	Site:
HIV Status:		
HIV +ve (pos) ☐	HIV –ve (neg) ☐	HIV Status Unknown ☐

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Guidelines for the Management of Tuberculosis in the Jamaican Population

MINISTRY OF HEALTH and WELLNESS, JAMAICA

TUBERCULOSIS INVESTIGATION FORM

Section 6 – Home and Contact Investigation (To be completed within 6 weeks of initial notification)

Name of Case: _____

Investigating Officer: _____

Facility Name: _____

Medical Record No: _____

Name	Age	Sex	Contacts (Household and relationship)	Address	Symptoms – Cough, fever, night sweats, weight loss – specify	Previo us BCG	Mantoux Date	Mantoux Reading	X-ray Date	X-ray Findings	Contact Classification Suspected TB
1.		M ☐ F ☐			Cough Y ☐ N ☐ Fever Y ☐ N ☐ Night Sweats Y ☐ N ☐ Weight Loss Y ☐ N ☐	Y ☐ N ☐					Y ☐ N ☐
2.		M ☐ F ☐			Cough Y ☐ N ☐ Fever Y ☐ N ☐ Night Sweats Y ☐ N ☐ Weight Loss Y ☐ N ☐	Y ☐ N ☐					Y ☐ N ☐
3.		M ☐ F ☐			Cough Y ☐ N ☐ Fever Y ☐ N ☐ Night Sweats Y ☐ N ☐ Weight Loss Y ☐ N ☐	Y ☐ N ☐					Y ☐ N ☐

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Guidelines for the Management of Tuberculosis in the Jamaican Population

Have there been confirmed TB case(s) in family or household? If yes, list them below

Name	Relationship to case	Date of illness	Treated	Comments
			Y ☐ N ☐ U ☐	
			Y ☐ N ☐ U ☐	
			Y ☐ N ☐ U ☐	

Population of Household: _____ **Number of rooms in dwelling:** _____ **Number of persons per room:** _____

Ventilation:	Cleanliness:	Water Supply:	Milk supply:	Comments:
Good ☐ Fair ☐ Poor ☐	Good ☐ Fair ☐ Poor ☐	Good ☐ Fair ☐ Poor ☐	Pasteurized ☐ Non-pasteurized ☐	

Additional Information/Action Taken: _____ Hypothesis as to source: _____

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Guidelines for the Management of Tuberculosis in the Jamaican Population

MINISTRY OF HEALTH and WELLNESS, JAMAICA

TUBERCULOSIS INVESTIGATION FORM

Section 7 to be completed and submitted to the parish MO(H) when case has completed treatment

Name of Case: _____ Date of Birth: _____

Medical Record No: _____ Investigating Officer: _____

Section 7 – Treatment continuation and termination

Continuation phase:	Date Started	Date completed:
Supervision method:	Name of DOTs Supervisor:	
Drug regime:	Dosage:	Comments:
☐ RI		
☐ Second line drugs		
Name of Drugs:		
☐ Third line drugs/injectables		

Follow-up Laboratory investigation

X-ray:	Date:	Pos/Neg:	Findings:	Comments:
2 –month check		Smear: Pos ☐ Neg ☐	Culture: Pos ☐ Neg ☐	
4/5-month check		Smear: Pos ☐ Neg ☐	Culture: Pos ☐ Neg ☐	
6/8-month check		Smear: Pos ☐ Neg ☐	Culture: Pos ☐ Neg ☐	
Repeat X-Ray		Results:		
Other Test (Specify):		Results:		
Other Test (Specify):		Results:		

Date Investigation Completed: _____

Name of Investigator(s): _____

Signature: _____

Name of Investigator(s): _____

Signature: _____

Parish MO(H): _____

Signature: _____

Parish MO(H) Comment: _____

_____ Date: _____

Regional Epidemiologist/RTD: _____

Signature: _____

Regional Epidemiologist/RTD Comment: _____

_____ Date: _____

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Section 8 –Treatment Outcome Classification

To be completed by the Ministry of Health and Wellness

Classification <i>(see definitions below)</i>	Date <i>(dd/mm/yyyy)</i>	Comments
Cured Y ☐ N ☐	Date of first negative lab result: Date of last negative lab result:	
Treatment completed Y ☐ N ☐	Date treatment completed:	
Treatment Failed Y ☐ N ☐	Date:	
Died Y ☐ N ☐	Date of Death:	
Lost to Follow-up Y ☐ N ☐	Date last seen:	
Not Evaluated Y ☐ N ☐	Date last seen:	
MDR Y ☐ N ☐	Date:	

Additional Comments:

Date received at HIV/STI/TB Unit: _____

National TB Coordinator: _____ Signature: _____

National TB Coordinator Comment:

_____ Date: _____

Date received at National Surveillance Unit: _____

NSU Officer: _____ Signature: _____

NSU Comment: _____ Date: _____

Ministry of Health and Wellness, Jamaica

Tuberculosis Investigation Form

Updated October 2019

Name:

Medical Record No:

Key: Y = Yes; N = No; U= Unknown

Definitions for Tuberculosis

Treatment History

New case: A patient who has never been treated for TB or who has taken anti-TB drugs for less than one month.

Previously treated patients: Patients who have received 1 month or more of anti-Tb drugs in the past.

- i. Relapse patients:** A patient previously treated for TB who was declared cured or treatment completed at the end of their most recent course of treatment, and are now diagnosed with a recurrent episode of TB.
- ii. Treatment after failure patients:** A patient who has been previously treated for Tb and whose treatment failed at the end of their most recent course of treatment.
- iii. Treatment after loss to follow up patients:** A patient who has been previously treated for TB and were declared loss to follow-up at the end of their most recent course of treatment (previously known as treatment after default patients).
- iv. Other previously treated patients:** Patients who have been previously treated for TB but whose outcome after their most recent course of treatment is unknown or undocumented

Definitions of treatment outcomes for drug-susceptible TB

- i. Cured:** A pulmonary TB patient with bacteriologically confirmed TB at the beginning of treatment who has was smear or culture negative in the last month of treatment and on at least one previous occasion.
- ii. Treatment completed:** A TB patient who completed treatment without evidence of failure, but with no record to show that sputum smear or culture results in the last month of treatment and on at least one previous occasion were negative, either because tests were not done or because results are unavailable.
- iii. Died:** A TB patient who dies for any reason before starting or during the course of treatment.
- iv. Treatment failed:** A TB patient whose sputum or culture is positive at month 5 or later during treatment.
- v. Loss to follow-up:** A TB patient who did not start treatment or whose treatment was interrupted for 2 consecutive months or more.
- vi. Not evaluated:** A TB patient for whom no treatment outcome is assigned. This includes cases “transferred out” to another treatment unit as well as cases for whom the treatment outcome is unknown to the reporting unit.
- vii. Treatment success:** A TB patient who is cured or who has completed treatment.

APPENDIX II

PAEDIATRIC EXTRA-PULMONARY TUBERCULOSIS CLINICAL PRESENTATION AND MANAGEMENT

Site of EPTB	Typical Clinical Presentation	Investigation	Comment
TB Adenitis	Asymmetrical, painless, non-tender lymph node enlargement for more than one month +/- discharging sinus. Most commonly in neck area.	Fine needle aspiration when possible for culture and histology. TST usually positive - not necessary for diagnosis.	Treat if axillary node enlarged on same side as BCG, consider BCG disease.
Pleural TB	Dullness on percussion and reduced breath sounds +/- chest pain	CXR Pleural tap	Treat if pus in pleural tap, consider empyema
Usually young (< 5 years) with disseminated disease and severely ill			
TB Meningitis	Headache, irritability/abnormal behaviour, vomiting (without diarrhoea), lethargic/reduced level of consciousness, convulsions, neck stiffness, bulging fontanelle, cranial nerve palsies.	Lumbar puncture to obtain CSF, CXR	Hospitalise for TB treatment
Miliary TB	Non-specific, lethargic, fever, wasted	CXR	Treat and refer
Usually 5 years and older			
Abdominal TB	Abdominal swelling with ascites or abdominal masses	Ascitic tap	Refer
Spinal TB	Deformity of spine May have lower limb weakness/paralysis	X-ray spine	Refer
Pericardial TB	Cardiac failure, distant heart sounds, apex beat difficult to palpate	CXR, Cardiac ultrasound, Pericardial tap	Refer
TB Bone and Joint	Swelling at end of long bones with limited movement. Unilateral effusion of usually knee or hip	X-ray bone/joint Joint tap#	Refer

APPENDIX III

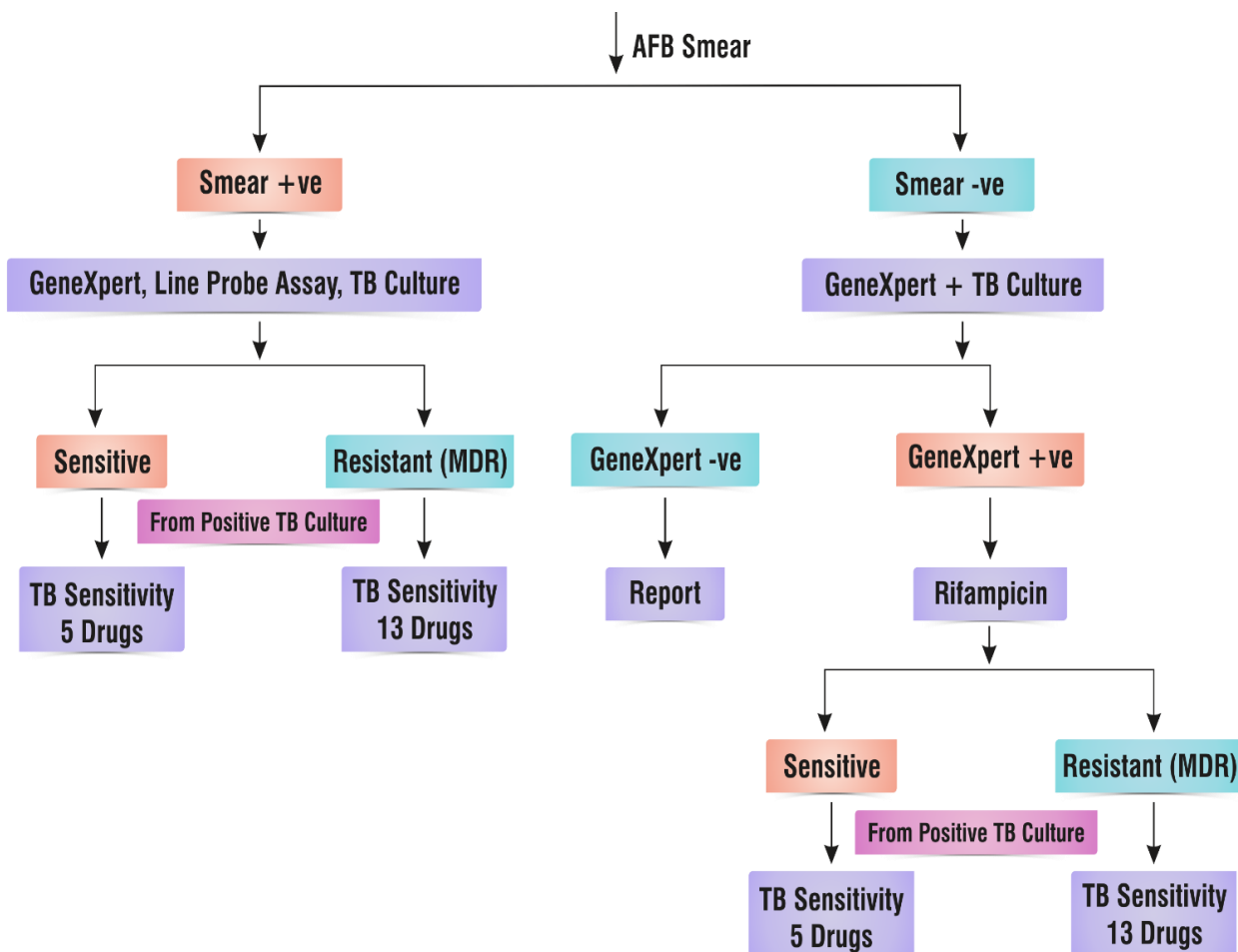
CLASS 1 NOTIFICATION FORM

CLASS 1 REPORTING FORM - INDIVIDUAL NOTIFICATION (ON SUSPICION)

Date of Report: ____ / ____ / ____ (DD/MM/YY)		NEW CASE / PREVIOUSLY REPORTED CASE (Circle One)	
Diagnosis: _____			
Case Demographic Information			
Name (including pet name): _____		Sex: ____	Age: ____ D.O.B. ____ / ____ / ____ (dd/mm/yy)
Address: (Include Landmark)	Lot #: ____ Street ____ (Name)	Street Type: ____ (Drive, Road, Close etc)	
Community	Neighbouring Community/District: ____		Parish: ____
Workplace/School:	Occupation: _____		
	History of overseas travel in past 4-6 weeks? Y / N		
(H) Phone #: _____	(Wk) Phone #: _____		
Name of NOK/Parent: _____		Relationship to case: _____	
Address of NOK/Parent: _____		Phone No.: _____	
Clinical Information:			
Symptoms:	Hosp./Facility Name: _____		
	Medical Record #: _____		
Date of onset: ____ / ____ / ____ (dd/mm/yy)	Case admitted to Hosp?: ____ / ____ / ____ (dd/mm/yy) Y / N (Circle one)		
Specimen Taken	Date of Admission: ____ / ____ / ____ (dd/mm/yy)		
Specimen Date: ____ / ____ / ____ (dd/mm/yy)	Ward: _____		
Result (s): _____	If dead, Date of Death: ____ / ____ / ____ (dd/mm/yy)		
Notifier Information			
Name of notifier: _____	Phone #: ____ / ____ / ____ (dd/mm/yy)		
Address: _____	Parish MO(H) Signature _____		
Comments: _____	Forwarded to R.S.O. ____ / ____ / ____ (dd/mm/yy)		
	Forwarded to Surveillance Unit ____ / ____ / ____ (dd/mm/yy)		

Ministry of Health, Surveillance Unit, July 2018

APPENDIX IV GENEXPERT INVESTIGATION



APPENDIX V
PHARMACOVIGILANCE FORM (AVAILABLE ON MOHW WEBSITE)

“PHARMWATCH”

MINISTRY OF HEALTH AND WELLNESS
DRUG MONITORING FORM

HEALTH CARE PROFESSIONALS

**ENSURE
SAFER**



PHARMACEUTICALS

**PARTICIPATE IN THE DRUG
MONITORING PROGRAMME**

**Report drug failure and adverse reactions with
medications and suspected counterfeit product**

An adverse reaction occurs when the patient outcome is:

Death, life-threatening (real risk of dying), hospitalization
(initial or prolonged), disability (significant, persistent or permanent),
congenital defect, permanent impairment, allergic reactions,
gastrointestinal distress.

Report even if:

- You're not certain whether the product caused the adverse reaction
- You don't have all the details

Who can report?

Any health care professional (Physician, Pharmacist, Dentist, Nurse)
Any patient who has experienced an adverse drug reaction

Where to report:

After completing, please return this form to: Dr. Maxine Gossell-Williams
Department of Basic Medical Sciences
Pharmacology Section, University of the West Indies
Tel: 927-2216; fax 977-3823
Email: maxine.gossell@uwimona.edu.jm

or

Mrs. Valerie Germain
Standards & Regulation Division
Ministry of Health
45-47 Barbados Ave, Kingston 5
Tel: 633-7150; fax: 630-3630
Email: germainv@moh.gov.jm

or

Ms. Angelika Steiner
Standards & Regulation Division
Ministry of Health
45-47 Barbados Ave, Kingston 5
Tel: 633-7148 fax: 630-3630
Email: steinera@moh.gov.jm

For additional information or for reporting online please visit the Ministry of Health and Wellness's website at
www.moh.gov.jm

What happens when the Form is submitted?

Any information provided in this form will be handled confidentially. The identities of the health care professional, patient or any other person reporting will be held in strict confidence and protected to the fullest extent. All reports will be assessed and causality analysis decided by Ministry of Health in due course. It is the ultimate responsibility of MOH to decide how to act on the information. It is also the responsibility of the Ministry to decide whether the incidences of reports will require further evaluation of drug performance. The Ministry will further provide the relevant pharmaceutical company with a summary of its findings and subsequent decision regarding intervention.

“PharmWatch” is a collaborative effort between the Ministry of Health and Wellness and the Pharmacology Section of the University of the West Indies.

“PHARMWATCH” DRUG MONITORING FORM

Prepared by: Maxine Gossell-Williams

Department of Basic Medical

Sciences Pharmacology Section

A. PATIENT DETAILS

1. Patient Initials: (First, Last)	3. Gender: ☐ M ☐ F	3. Date of Birth: (mm/dd/yyyy)	4. Ethnicity	5. Weight: (Kg)	6. Height: (cm)
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B. SUSPECTED DRUG EVENT

7. Outcomes attributed to use of drug (check all that apply): ☐ Failure of therapy ☐ Allergy ☐ Disability ☐ Life threatening ☐ Hospitalisation ☐ Death _____ mm/dd/yyyy ☐ Other (describe) _____	8. Describe event or problem	9. Date event started (mm/dd/yyyy)
		10. Date event ended (mm/dd/yyyy)

Guidelines for the Management of Tuberculosis in the Jamaican Population

11. Describe action taken in response (e.g. drug changed, prolonged-therapy, increased dose)	12. Describe other relevant history including abnormal laboratory test results, days of hospitalization.
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C. DRUG INFORMATION

13. Name of suspected drug (give specific name on package)	14. Dose & Route	15. Indication	16. Batch number if known
17. Name of other drugs taken (give specific name on package)	18. Dose & Route	19. Indication	20. Batch number if known

D. REPORTING HEALTH PROFESSIONAL INFORMATION

21. Profession: _____ 22. Name: _____ 23. Address: _____	24. Telephone: _____ 25. Fax: _____ 26. Email _____
27. Also reported to: _____	
Signature	Date (mm/dd/yyyy)

FOR OFFICIAL USE ONLY

Received by: _____ Date received: _____	Action taken: _____

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